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Original Paper

Quasi-optimal Sampling from Gibbs States via Non-commutative Optimal Transport Metrics

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Abstract. We study the problem of sampling from and preparing quantum Gibbs states of local commuting Hamiltonians on hypercubic lattices of arbitrary dimension. We prove that any such Gibbs state which satisfies a clustering condition that we coin decay of matrix-valued quantum conditional mutual information (MCMI) can be quasi-optimally prepared on a quantum computer. We do this by controlling the mixing time of the corresponding Davies evolution in a normalized quantum Wasserstein distance of order one. To the best of our knowledge, this is the first time that such a non-commutative transport metric has been used in the study of quantum dynamics, and the first time quasi-rapid mixing is implied by solely an explicit clustering condition. Our result is based on a weak approximate tensorization and a weak modified logarithmic Sobolev inequality for such systems, as well as a new general weak transport cost inequality. If we furthermore assume a constraint on the local gap of the thermalizing dynamics, we obtain rapid mixing in trace distance for interactions beyond the range of two, thereby extending the state-of-the-art results that only cover the nearest neighbour case. We conclude by showing that systems that admit effective local Hamiltonians, like quantum CSS codes at high temperature, satisfy this MCMI decay and can thus be efficiently prepared and sampled from.

Contents

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1. Introduction

The problem of Gibbs state preparation is fundamental in statistical mechanics. Given a Hamiltonian H on a quantum system and an inverse temperature $\beta < \infty$, the Gibbs state $\sigma^\beta = e^{-\beta H}/\mathrm{Tr}[e^{-\beta H}]$ describes the properties of the system at its thermal equilibrium [7]. Thus, Gibbs states and their fundamental properties are essential for the study of quantum systems, for example in the contexts of simulation of many-body systems [73], their use as topological quantum memories [66] or their thermalization processes [70,78].

Gibbs sampling has been for a long time a cornerstone in statistical mechanics. It is an extremely relevant problem for the fields of physics and computer science, in both the context of classical and quantum mechanics, in part due to its applications in multiple scenarios [55,88]. Classical Markov chain Monte Carlo (MCMC) methods constitute one of the canonical tools for sampling from classical Gibbs states of spin systems [68]. These methods have been demonstrated to be efficient at sufficiently high temperatures [71] and are widely considered efficient in practice [17]. However, there is no clear consensus on how to best extend such algorithms to quantum Gibbs sampling. Many relevant classical algorithms for Gibbs sampling have appeared in the past few years [3,4,8,25]. Here we take a further step and aim at proving the efficiency of quantum extensions thereof.

A good Gibbs sampler is expected to prepare the Gibbs state in polynomial time. In the past few years, many quantum algorithms inspired by the classical Monte Carlo have been proposed in [80,86,89], among others, but for long without any provable guarantee or only validated under strong theoretical assumptions [22,82]. This has drastically changed with the very recent appearance of [26,27], where a quantum algorithm to prepare quantum Gibbs states was proposed and subsequently shown to be efficient in [76]. This has inspired an important collection of works in the context of quantum Gibbs sampling, with algorithms not only based on dissipation [11,26,39,48,80], but also in some other techniques [52,57,64]. A detailed discussion on some relevant literature for quantum Gibbs sampling is presented in Sect. 3.

In this paper, we investigate quantum Gibbs sampling with Davies generators [37,38] associated with local commuting Hamiltonians. Local commuting Hamiltonians hereby constitute a class of systems that non-trivially extends beyond the classical regime [1,5,18,53] and include highly entangled systems like CSS codes (e.g. the Toric code) and quantum double models [34,61,83]. Davies Lindbladians constitute the canonical objects to model the Markovian dissipation of a quantum system weakly coupled with an infinite-dimensional thermal bath and can be regarded as the natural quantum analogue of Glauber dynamics [71]. Hence, they represent a natural tool for quantum Gibbs sampling. Many recent efforts to develop efficient Gibbs samplers have been undertaken in the context of commuting Hamiltonians, such as for 1D translation-invariant systems at any positive temperature [10,11,59] and high-dimensional systems at high-enough temperature [33,59]. In high dimensions at low temperature, efficient Gibbs samplers are only known to exist for Kitaev's quantum double models in 2D [2,40,63,67].

The efficiency of Gibbs samplers is determined by the speed of convergence, or mixing, of the corresponding Lindbladian towards its thermal equilibrium, estimated through the notion of *mixing time*. Generically, the algorithmic sample complexity can be upper bounded by the system size times its mixing time [69,80]. Both classically and quantumly, the most frequent way of estimating this mixing time is through the spectral gap of the Lindbladian [9,60]. A positive spectral gap provides an upper bound on the mixing time that scales linearly with the system size, in a regime known in the literature as *fast mixing*, or *polytime mixing* [53], but this is generally believed to be an overestimation. A more precise estimate follows from the existence of a $\Omega(1/\operatorname{poly}\log N)$ modified logarithmic Sobolev inequality [65], which implies a mixing time that scales polylogarithmically with the system size, a regime known as *rapid mixing*. For continuous time local reversible Glauber dynamics on bounded-degree graphs, it has been shown that the mixing time obeys $\Omega(\log N)$ [56, Theorem 4.1] while in [30] the bound was achieved for a subclass of models, establishing the optimality of rapid mixing. In this regime, the algorithmic sample complexity scales linearly with system size, up to a polylogarithmic correction, a property commonly referred to as *optimal sampling*.

In accordance with this nomenclature, we use this manuscript the term quasi-rapid mixing to describe a mixing time that scales quasi-logarithmically with system size, specifically $\mathcal{O}(\exp\operatorname{poly}\log\log)$. Such a mixing time implies hence an algorithmic complexity that remains linear up to quasi-logarithmic corrections—significantly better than the quadratic scaling one would obtain from a naive gap assumption. In line with the literature, we refer to this preparation complexity as quasi-optimal [56]. In terms of scaling, however, it is much closer to the linear scaling—optimal for preparation circuit complexity—than to any polynomial growth. Similarly, a quasi-logarithmic mixing time can be considered quasi-optimal in the sense that it is far closer to the polylogarithmic scaling, which is optimal, than to any polynomial scaling.

Whereas some quantum Gibbs samplers associated with non-commuting Hamiltonians were previously only known to exhibit fast mixing [76], recent results have established that rapid mixing can also occur in this setting under the existence of Lieb–Robinson bounds [77]. Until now, examples of rapid mixing had been primarily found in the context of commuting Hamiltonians [10,11,33,59]. This new development expands the scope of rapid mixing beyond commuting systems, though our focus in this manuscript remains on the latter.

2. Main Results

In this work, we demonstrate the quasi-rapid mixing in normalized Wasserstein distance of Davies generators associated with local commuting Hamiltonians, assuming that their Gibbs state satisfies a specific clustering condition. By additionally assuming a polynomially decaying gap of the local Davies generator, we strengthen our result to rapid mixing in trace distance.

To our knowledge, this is the first instance where quasi-rapid mixing is derived solely from a static and explicit notion of decay of correlations in the steady-state or Gibbs state. Furthermore, while mixing times based on normalized Wasserstein distances have been studied in the classical setting [8], we believe this is the first time that mixing with respect to normalized quantum Wasserstein distances has been considered in the quantum setting, thereby further connecting quantum optimal transport theory and Gibbs sampling [16,41–43,75].

We introduce the *matrix-valued quantum conditional mutual information (MCMI)* in Definition 2.3, whose decay is more general than that of both the covariance and the conditional mutual information—the former shown to be sufficient for rapid thermalization in one-dimensional lattices [59]. Our first main result requires only the uniform decay of the MCMI across four arbitrary partitions of the underlying lattice:

Theorem 2.1 (Quasi-optimal preparation of MCMI-decaying Gibbs states, informal). *Let σ be a Gibbs state of a local commuting Hamiltonian that satisfies uniform exponential decay of its MCMI. Then there exists a quantum circuit with circuit complexity and runtime*

$$\mathcal{O}\left(N\,\text{quasi-log}(N), \text{quasi-poly}\left(\frac{1}{\epsilon}\right)\right) = o(N^{1+\delta})$$

that prepares a state ρ that is ϵ-close in normalized W_1 distance to σ, for any $\delta > 0$.

Corollary 2.2. *Let σ be a Gibbs state of a commuting Hamiltonian which satisfies the marginal commuting property, see Appendix A.4.3, at uniformly high temperature. Then it can be prepared quasi-optimally in normalized Wasserstein distance, i.e. there exists a quantum circuit with complexity and runtime bounded by*

$$\mathcal{O}\left(N\,\text{quasi-log}(N), \text{quasi-poly}\left(\frac{1}{\epsilon}\right)\right) = o(N^{1+\delta})$$

that prepares a state ρ that is ϵ-close in normalized W_1 distance to σ, for any $\delta > 0$.

To establish the quasi-rapid mixing time and achieve efficient sampling, we prove a suitable *weak modified logarithmic Sobolev inequality* (weak MLSI) for the Davies evolution corresponding to a local commuting Hamiltonian (see Corollary 2.13). This yields exponential decay of the relative entropy between any time-evolved initial state and the Gibbs state, with a prefactor depending polynomially on the system size. Following recent work [11, 33, 59], we employ a "divide-and-conquer" strategy, using the uniform decay of the MCMI for the global Gibbs state and leveraging the locality of the Davies generators to estimate global mixing in terms of local mixing plus an additive error term.

More precisely, we employ the uniform decay of the MCMI to prove a general entropy factorization (Lemma 2.6), which we extend to an approximate tensorization on $\mathbb{Z}^D$ (Theorem 2.8). This result reduces the relative entropy of two states supported on the global lattice to a sum of conditional relative entropies in smaller subregions, with an additive error controlled by the decay of the MCMI. This is the crucial ingredient to prove the weak MLSI (Corollary 2.13). Finally, to apply this weak MLSI to our Wasserstein distance-based mixing time, we derive a general *weak transport cost* inequality (Theorem 2.11).

Notably, unlike previous results on MLSIs of Davies dynamics in many-body quantum spin systems [10, 11, 59], our main result (Theorem 2.10, see also Theorem C.1), shows that quasi-rapid mixing is implied solely by the decay

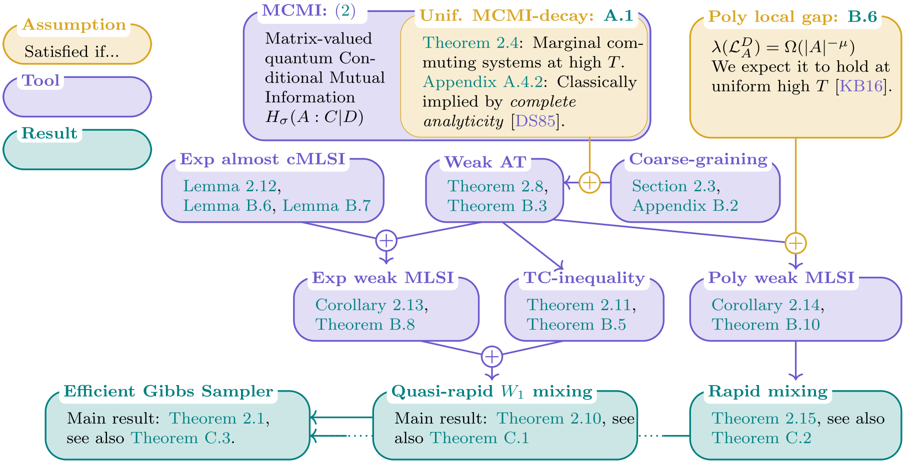

FIGURE 1. Logical structure of the results in this article. In orange, we represent the assumptions considered; in purple, the main technical lemmas, of independent interest; and in green, the main technical results regarding the mixing times of the evolution and their application in the context of Gibbs sampling. Note that the main result is implied by either the quasi-rapid W_1 mixing, or the rapid mixing

of the MCMI, without additional assumptions on quantities like the local gap. This is significant because it is the first time that a dynamical property as strong as quasi-rapid mixing can be derived for physically relevant quantum spin systems solely from a static condition on the system's equilibrium (the Gibbs state), without further weaker dynamical assumptions.

Moreover, under further assumptions—specifically, an at most polynomially decaying gap of the local Davies generators, which we expect to hold at uniformly high temperature—we also show rapid mixing of these dynamics with respect to the trace distance (see Theorem 2.15).

In the following sections, we provide a detailed overview of these results and the required lemmas, while omitting most technical details and proofs. A formal introduction to the necessary mathematical concepts, along with the full proofs, can be found in Appendices A–C. We conclude with Appendix D, where we present examples of systems for which the required decay of correlations—specifically, the decay of the MCMI—holds, making our main results applicable.

For the readers' convenience, we collect all main results, important technical lemmas and assumptions in Fig. 1.

2.1. Notation and Preliminaries

This work considers quantum spin systems on finite hypercubic lattices $\Lambda \subset \mathbb{Z}^D$. The finite-dimensional Hilbert space of said systems is $\mathcal{H}_\Lambda := \bigotimes_{x\in\Lambda} \mathcal{H}_x$, where each site contains a qudit $\mathcal{H}_x \simeq \mathbb{C}^d$. The operator norm on the set of bounded operators $\mathcal{B}(\mathcal{H})$ is denoted as $\|\cdot\|_\infty$ and the trace norm as $\|\cdot\|_1$. The set of all quantum states, i.e. non-negative, trace-normalized operators on $\mathcal{H}$, is denoted as $\mathcal{S}(\mathcal{H})$. The partial trace is $\operatorname{tr}_A : \mathcal{B}(\mathcal{H}_\Lambda) \to \mathcal{B}(\mathcal{H}_{\overline{A}})$, where $\overline{A} := \Lambda \setminus A$ is the complement region of $A \subset \Lambda$. A $(\kappa, r)-$local, commuting, $J-$bounded Hamiltonian $H_\Lambda \in \mathcal{B}(\mathcal{H})$ is a self-adjoint operator $H_\Lambda = \sum_{A \subseteq \Lambda} h_A$, such that $||h_A||_\infty \leq J$, $h_A = 0$, whenever $|A| > \kappa$, or $\operatorname{diam}(A) > r$, and for each $A, B \subset \Lambda$, $[h_A, h_B] = 0$, where $[X, Y] := XY - YX$ is the commutator of two operators $X, Y \in \mathcal{B}(\mathcal{H})$. We define the boundary of a sublattice $R \subseteq \Lambda$, tied to the connectivity of the Hamiltonian as $\partial R := \{k \in \Lambda : \exists A \subseteq \Lambda, h_A \neq 0, A \cap R \neq \emptyset, A \cap \overline{R} \neq \emptyset, k \in A\}$ and write $R\partial := R \cup \partial R$. For a sublattice $R \subseteq \Lambda$, we define the *local Hamiltonian* as $H_R := \sum_{A \subseteq R} h_A$ and set the *growth constant* g of the system (Λ, H_Λ) as $g := \max_{k \in \Lambda} |\{A \subseteq \Lambda : h_A \neq 0, k \in A\}|$. Hence, it satisfies $||H_R||_\infty \leq gJ|R|$. The local, resp. global Gibbs state at inverse temperature $\beta > 0$, $R \subset \Lambda$, resp. $R = \Lambda$ is given by $\sigma^R := \frac{e^{-\beta H_R}}{\operatorname{Tr}[e^{-\beta H_R}]}$ and its reduced state is defined as $\sigma_R := \operatorname{tr}_{\overline{R}}[\sigma^\Lambda]$. The relative entropy between a state ρ and a full-rank state $\sigma > 0$ is defined as $D(\rho\|\sigma) := \operatorname{tr}[\rho(\log \rho - \log \sigma)]$. The generator of the semigroup dynamic we are going to investigate in this paper is called the Davies semigroup, which for a fixed local commuting Hamiltonian H_Λ and inverse temperature β is given by

$$\mathcal{L}_\Lambda^D(\rho) := \sum_{k \in \Lambda} \mathcal{L}_k^D \quad \text{with} \quad \mathcal{L}_k^D(\rho) := \sum_{\alpha, \omega} \chi_{\alpha,k}^{\beta,\omega} \left(S_{\alpha,k}^\omega \rho S_{\alpha,k}^{\omega,\dagger} - \frac{1}{2} \left\{ \rho, S_{\alpha,k}^{\omega,\dagger} S_{\alpha,k}^\omega \right\} \right). \tag{1}$$

The $S_{\alpha,k}^\omega$ depend on H_Λ through $e^{itH_\Lambda} S_{\alpha,k} e^{-itH_\Lambda} = \sum_\omega e^{it\omega} S_{\alpha,k}^\omega$ for all $t \in \mathbb{R}$, where $\{S_{\alpha,k}\}_\alpha$ labels a set of self-adjoint operators supported on k that form a Kraus decomposition of the partial trace tr_k. The prefactors $\chi_{\alpha,k}^{\beta,\omega}$ satisfy the KMS condition $\chi_{\alpha,k}^{\beta,-\omega} = e^{-\beta\omega} \chi_{\alpha,k}^{\beta,\omega}$ and we assume them to be uniformly bounded from above and below as $0 < \chi_{\min}^\beta \leq \chi_{\alpha,k}^{\beta,\omega} \leq \chi_{\max}^\beta$. We define the local Davies generators as $\mathcal{L}_A^D := \sum_{k \in A} \mathcal{L}_k^D$ and note that for all $A \subseteq \Lambda$ the generated dynamics converge to projecting channels, we denote by $E_A = \lim_{t \to \infty} e^{t\mathcal{L}_A}$.

2.2. Uniform Decay of the MCMI

In classical spin systems, there is a strong connection between static correlation properties in the thermal or steady state—such as spectral independence [6] or decay of correlations [71]—and the mixing time of the thermalizing dynamics. Specifically, Glauber dynamics, which models the thermalization of a discrete spin system, rapidly approaches the equilibrium Gibbs measure if and only if correlations between spatially separated regions decay exponentially with the distance between them. A similar connection has been established in the quantum case, where notions of decay of correlations have been shown to be closely related to mixing [14,33,59]. This work builds on these previous

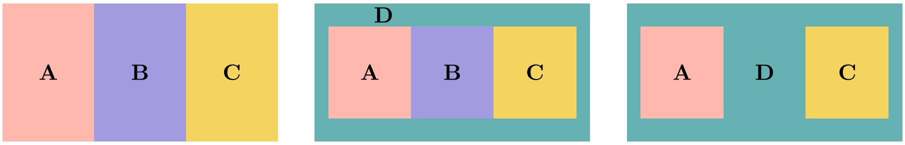

FIGURE 2. A lattice Λ is partitioned into four distinct regions, such that A and C are separated by either B or D or both. B is a system that is averaged over (through a partial trace) while D we condition on

results by introducing the *matrix-valued quantum conditional mutual information (MCMI)* as a correlation measure closely tied to the mixing properties of the so-called Davies dynamics. This quantity, previously considered in the study of quantum many-body systems [62], is closely related to quantum conditional mutual information and the mixing condition studied in [11,13,14], where the mixing property was identified as a condition to imply rapid mixing of Davies dynamics on 1D lattices.

Definition 2.3. (*Matrix-valued quantum conditional mutual information (MCMI)*) Given a quantum state $\sigma \in \mathcal{S}(\mathcal{H}_{ABCD})$ on some tensor product Hilbert space $\mathcal{H}_{ABCD} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C \otimes \mathcal{H}_D$, then its MCMI is defined as

$$\mathbf{H}_\sigma(A : C|D) := \log \sigma_{ACD} + \log \sigma_D - \log \sigma_{AD} - \log \sigma_{CD}, \tag{2}$$

and we will frequently write $H_\sigma(A : C|D) := ||\mathbf{H}_\sigma(A : C|D)||_\infty$ for its operator norm (Fig. 2).

It is a general notion of mixing that controls both:

1. the conditional mutual information $I_\sigma(A : C|D) := \operatorname{tr}[\sigma \mathbf{H}_\sigma(A : C|D)] \leq H_\sigma(A : C|D)$;
2. the mutual information and thus also covariance $I_\sigma(A : C) := \operatorname{tr}[\sigma_{AC} \mathbf{H}_\sigma(A : C|\emptyset)] \leq H_\sigma(A : C|\emptyset)$

For σ a Gibbs state of a local Hamiltonian at inverse temperature β, the norm of the MCMI is believed to decay exponentially in $\operatorname{dist}(A, C)$ at high-enough temperature. This was, erroneously, thought proved in [62] for some time. Although we do not resolve this open question, we prove that the conjecture holds for *marginal commuting* Hamiltonians: We say that $H_\Lambda = \sum_{A\subseteq\Lambda} h_A$ is *marginal commuting* (see [14, Definition 3.5]) if the algebra $\mathcal{A}$ generated by $\{h_A\}_{A\subset\Lambda}$ is commuting and closed under arbitrary partial traces, i.e. for all $A \subseteq \Lambda$, $I_A \otimes \operatorname{tr}_A[\mathcal{A}] \subseteq \mathcal{A}$. This condition, at high-enough temperature, is a particular case of applicability of the following theorem:

Theorem 2.4 (Uniform decay of the MCMI). *Let σ be the Gibbs state to some (κ, r)-local and J-bounded Hamiltonian on a graph Λ that has growth constant g, then there exist some constants $K, \xi > 0$ such that for any partition $\Lambda = A \sqcup B \sqcup C \sqcup D$*

$$H_\sigma(A : C|D) \leq K|\Lambda| \exp(-\operatorname{dist}(A, C)/\xi), \tag{3}$$

if H_Λ admits a strong local effective Hamiltonian in the sense of Appendix A.4.3, see also Definition 3.1 of [14]*. In this case, we say that the state σ satisfies* uniform decay of the MCMI*.*

Remark 1. In [14], it was shown that all marginal commuting systems at suitably high temperature satisfy this condition and hence, by the above theorem, also uniform MCMI decay, see Appendix D.2 and Theorem D.2. These notably include all quantum CSS codes and all classical Hamiltonians with a Gibbs state that satisfies complete analyticity [23, 44, 45], see Appendix A.4.2.

2.3. Entropy Factorization and a Weak AT

Statements on (strong) approximate tensorization (AT) relate conditional relative entropies on lattices to ones on subsystems with multiplicative error terms that quantify the closeness of the fixed point to being a product state. Such statements are central in both the classical and quantum setting when proving MLSIs and efficient bounds on mixing times of certain dynamics. The issue, however, is that, unlike in the classical setting, so far no general iterable approximate tensorization statement is known in the quantum setting. The ones that are known are sufficient when considering the fixed point to be a tensor product [29], systems on one-dimensional lattices [10, 11], or nearest neighbour interactions [33, 59] but do not allow to go beyond these settings.[1] In this work, we circumvent this problem by considering a *weak approximate tensorization (weak AT)* which will suffice for the quasi-rapid mixing of the Davies dynamics in W_1-distance.

To state our weak AT, we first want to introduce the concept of conditional relative entropy which will serve as a useful tractable proxy for the Davies conditional relative entropy we ultimately wish to study. For more detail, see Appendix A.3 and (50).

Definition 2.5 (*Conditional relative entropy*[29]). Given a lattice Λ, the conditional relative entropy on $A \subset \Lambda$ between two states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_\Lambda)$ is defined as

$$D_A(\rho\|\sigma) := D(\rho\|\sigma) - D(\rho_{\overline{A}}\|\sigma_{\overline{A}}). \tag{4}$$

We are now able to show the following novel relation between the conditional relative entropies:

Lemma 2.6 (Weak entropy factorization). *Let $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C \otimes \mathcal{H}_D$ and $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ with $\sigma > 0$, then*

$$D_{ABC}(\rho\|\sigma) \leq D_{AB}(\rho\|\sigma) + D_{BC}(\rho\|\sigma) + \left|\left|\mathbf{H}_\sigma(A : C|D)\right|\right|_\infty . \tag{5}$$

[1] Note that the nearest neighbour interaction results in [33, 59] are assumed on two colourable graphs. Hence, they do not simply hold for grouped together interactions, i.e. extend to more than nearest neighbour interactions on general graphs. More specifically, given, for example, a two-dimensional square lattice with three local interactions, it is not possible to group the interactions into another two colourable graphs with nearest neighbour interactions. See also the discussion in [59] on page 40.

The proof only requires the data processing inequality and Hölder inequality for Schatten-p norms and can be found in Lemma B.1. Unlike the previously known entropy factorizations [29,33], our new results allow us to have a system D that we carry as a conditioning through the estimate which is key in the iteration of this statement, yielding a weak AT statement for lattices of arbitrary dimension later on. Note that classically (i.e. when all involved states commute) one can derive a multiplicative correction, i.e. a strong *entropy factorization* of the form

$$(1 - ||\exp(\mathbf{H}_\sigma(A:C|D)) - I||_\infty) D_{ABC}(\rho\|\sigma) \leq D_{AB}(\rho\|\sigma) + D_{BC}(\rho\|\sigma)\,. \tag{6}$$

Establishing a similar relation in the quantum setting would allow us to lift all weak inequalities to their strong counterparts making for example the requirement of uniform polynomial local gap as a prior for rapid mixing obsolete (c.f. Theorem 2.15). However, establishing such an inequality for quantum systems where $D \neq \emptyset$ remains still open.

To extend the above entropy factorization to a (weak) approximate tensorization statement on a lattice $\Lambda_L := [\![-L, L]\!]^D \subset \mathbb{Z}^D$, we require a suitable *coarse-graining* of Λ w.r.t. which we split the conditional relative entropy. For $D = 1, 2$ such were given in [19]. Effectively a suitable coarse-graining is a family of subsets $\{C_{a,i}\}_{a,i}$ that cover Λ, are all suitably local, and have overlaps between neighbouring levels a.

Lemma 2.7. *For any $D \in \mathbb{N}$, there exists a suitable coarse-graining. Explicitly, given $r \leq k, c, \ell \leq 2L+1$ such that $\ell > 2D(k+c)$, there exists a coarse-graining $\{C_{a,i}\}_{a\in[\![D]\!],i}$ of $\Lambda_L = [\![-L, L]\!]^D$ made up of $D+1$ different levels indexed by a, such that*

1. *$\bigcup_{a,i} C_{a,i} = \Lambda$ and each site $x \in \Lambda$ is included in at most $D+1$ sets $C_{a,i}$,*
2. *$\{C_{a,i}\}_i$ is a collection of mutually disjoint subsets $\forall a \in [\![D]\!]$,*
3. *$|C_{a,i}| \leq \ell^D$,*
4. *Each level a has a suitable overlap c with the level coming before and after.*

An explicit construction of the covering may be found in Appendix B.2 with its key properties summarized in Lemma B.2 and a figure detailing the construction for the case $D = 2$ in Fig. 3.

With the entropy factorization as well as the lattice coarse-graining in place, we are now able to prove one of the core results of this paper, namely the weak approximate tensorization of Gibbs states of local commuting Hamiltonians that exhibit uniform decay of the MCMI.

Theorem 2.8 (Weak approximate tensorization). *Given a suitable coarse-graining of the lattices Λ with constants (k, c, l) as described above, and corresponding Davies channels $\{E_{C_{a,i}} : a \in [\![0, D]\!], i \in \mathcal{I}_a\}$ corresponding to the*

Gibbs state σ at temperature β of a local commuting Hamiltonian, the following inequality holds for all $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$:

$$D(\rho\|\sigma) \leq \sum_{a=0}^{D} \sum_{i_a \in \mathcal{I}_a} D(\rho\|E_{C_{a,i_a}}(\rho)) + \sum_{a=1}^{D} \zeta_a(\sigma),\ \zeta_a(\sigma) := ||\mathbf{H}_\sigma(X_a : Z_a|W_a)||_\infty\ , \tag{7}$$

where $W_a \sqcup X_a \sqcup Y_a \sqcup Z_a =: \overline{C^a} \sqcup \mathring{C}_a \sqcup (C_a \backslash \mathring{C}_a) \sqcup (C^a \backslash C_a)$ with $d(X_a, Z_a) = c \geq r$, for $a \in [\![1, D]\!]$. The explicit definition of these sets can be found in Appendix B.2. If σ further satisfies uniform decay of MCMI with constants K and ξ, it holds for any $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$ that

$$D(\rho\|\sigma) \leq \sum_{a=0}^{D} \sum_{i_a \in \mathcal{I}_a} D(\rho\|E_{C_{a,i_a}}(\rho)) + D2^D K|\Lambda|e^{-c/\xi}\,. \tag{8}$$

*We refer to (7) as a $(1, c_2)$-*weak AT *w.r.t. the coarse-graining $\{C_{a,i}\}_{a,i}$ where*

$$c_2 = \sum_{a=1}^{D} \zeta_a(\sigma) \leq K2^D K|\Lambda|e^{-\frac{c}{\xi}}\,.$$

The proof works by induction over a using the entropy factorization Lemma 2.6 and the coarse-graining introduced before, separating one level of the hierarchy in every step. It may be found in Appendix B.6.

2.4. Quasi-rapid Mixing of Lipschitz Observables

In this section, we prove the quasi-rapid mixing of the Davies dynamics w.r.t the quantum Wasserstein distance of order 1. We recall its definition. For composite systems $\mathcal{H}_\mathcal{I} = \bigotimes_{i\in\mathcal{I}} \mathcal{H}_i$, informally, W_1 quantifies a minimal "local change", which is enforced by a Lipschitz seminorm that measures how insensitive an observable can be made to each site by subtracting an operator acting on its complement. For further details and an example, we refer the reader to Appendix A.1.

Definition 2.9 (W_1-*distance from* [41]). For two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_\Lambda)$, the quantum Wasserstein distance of order 1 is given as the dual norm to a Lipschitz norm.

$$||\rho - \sigma||_{W_1} := \max_{H \in \mathcal{B}_{\mathrm{sa}}(\mathcal{H}_\mathcal{I}), ||H||_L \leq 1} \mathrm{tr}[H(\rho - \sigma)]\,, \tag{9}$$

where the Lipschitz norm of $H \in \mathcal{B}_{\mathrm{sa}}(\mathcal{H}_\Lambda)$ is defined as

$$||H||_L := 2 \max_{k\in\Lambda} \min_{\tilde{H} \in \mathcal{B}_{\mathrm{sa}}(\mathcal{H}_{\Lambda\backslash\{k\}})} \left|\left|H - I_k \otimes \tilde{H}\right|\right|_\infty\,. \tag{10}$$

One can show that for two states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_\Lambda)$, it satisfies $\frac{1}{2}\|\rho - \sigma\|_1 \leq \|\rho - \sigma\|_{W_1} \leq |\Lambda| \|\rho - \sigma\|_1$, and hence, it is an extensive norm that indicates how well two states differ locally [41].

Quite naturally we define the ϵ-mixing time for a primitive QMS with fixed point σ w.r.t this normalized quantum Wasserstein distance as

$$t_{\mathrm{mix}}^{W_1}(\epsilon) := \inf\left\{t \geq 0 : \forall \rho \in \mathcal{S}(\mathcal{H}_\Lambda),\ \left\|e^{t\mathcal{L}_\Lambda}(\rho) - \sigma\right\|_{W_1} \leq |\Lambda|\epsilon\right\}. \tag{11}$$

Note that mixing times based on normalized Wasserstein distances have also been considered in recent classical Gibbs sampling, see, for example, [8] in which the authors prove optimal sampling via algorithmic stochastic localization for certain spin glass models at high temperature in normalized Wasserstein-2 distance.

The rest of this section is dedicated to proving the first main result of this work:

Theorem 2.10 (Quasi-rapid Wasserstein mixing). *Let $\{\mathcal{L}^D_{\Lambda_L}\}_L$ be a family of Davies Lindbladians corresponding to a family of (κ, r)-local, commuting, J-bounded Hamiltonians $\{H_{\Lambda_L}\}_L$, each of which has growth constant g. Consider $\epsilon > 0$ and denote by $N = |\Lambda_L|$ the size of the lattice. Then, if the invariant states of $\mathcal{L}^D_{\Lambda_L}$, i.e. the Gibbs states σ^{Λ_L}, all satisfy uniform exponential decay of its matrix-valued quantum conditional mutual information (MCMI) as defined in (2), the semigroups $\{e^{t\mathcal{L}^D_{\Lambda_L}}\}_{t\geq 0}$ satisfy*

$$t^{W_1}_{\text{mix}}(\epsilon) = \text{quasi-poly}\left(\frac{1}{\epsilon^2}\,\text{poly}\log\frac{N}{\epsilon^2}\right) = \text{quasi-poly}(\epsilon^{-1})_{\epsilon\to 0}\text{quasi-log}(N)_{N\to\infty}. \tag{12}$$

For a precise upper bound on $t^{W_1}_{\text{mix}}(\epsilon)$, see Theorem C.1.

In the following, we give an overview of the main strategy and tools required to prove this result. The proof may be found in Appendix C.1 and consists of the following two ingredients.

1. Relating the W_1 distance to the relative entropy. We do this via a general *weak transport cost* (weak TC) inequality implied by the weak AT (7). This is an inequality relating the W_1 distance between any state and a fixed point of a QMS that satisfies a (weak) AT to their conditional relative entropy.

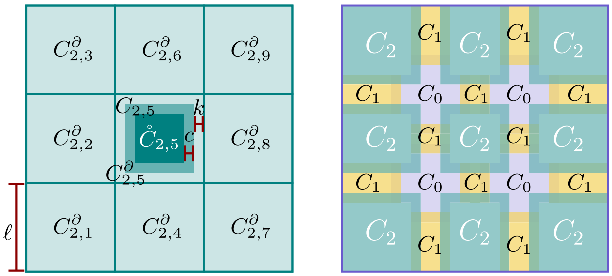

FIGURE 3. An example of the coarse-graining of a $D = 2$ dimensional lattice. On the left the largest regions in the top level $a = 2$ of the hierarchy with side length ℓ. On the right the different overlapping levels, labelled by $a = 2, 1, 0$, in different colours. Each of the local cells $C_{a,i}$ has size $|C_{a,i}| \leq \ell^2$ (Color figure online)

2. Showing sufficient decay of the relative entropy: We prove a (α, ϵ) weak MLSI, in which the cMLSI constant verifies $\alpha(\epsilon)^{-1} = \mathcal{O}(\exp \operatorname{poly}\log \frac{N}{\epsilon})$. This requires the combination of the (weak) AT with a cMLSI alike estimate relating the Davies conditional relative entropy $D(\rho \| E_A(\rho))$ to the entropy production of $\mathcal{L}^D_{A\partial}$ at an exponential cost in $|A\partial|$.

Let us begin with the statement of the weak transport cost inequality for Gibbs states of local commuting Hamiltonians:

Theorem 2.11 (Weak transport cost inequality). *Let σ be the Gibbs state at inverse temperature β of a local commuting Hamiltonian on a lattice Λ that satisfies a $(1, c_2)$-weak AT w.r.t. some coarse-graining $\{A_i\}_{i=1}^{n_A}$ of Λ, like in (7). Then it satisfies a (b_1, b_2)-*weak transport cost *inequality with $b_1 = 2\sqrt{2n_A}\max_i |A_i\partial|$ and $b_2 = |\Lambda|\sqrt{2c_2}$, i.e.:*

$$||\rho - \sigma||_{W_1} \leq \max_i 2\sqrt{2}|A_i\partial|\sqrt{n_A D(\rho\|\sigma)} + |\Lambda|\sqrt{2c_2} \tag{13}$$

The proof of this result may be found in Theorem B.5 of Appendix B.4. The second step estimates the Davies conditional relative entropy on some local region A with the entropy production on a slightly larger region:

Lemma 2.12. *Let $A \subseteq \Lambda$ and let $E_A := \lim_{t\to\infty} e^{t\mathcal{L}^D_A}$ be the conditional expectation of a local (on A) Davies generator at inverse temperature β corresponding to a local, commuting, $J-$bounded Hamiltonian that on the lattice Λ has growth constant g, then*

$$D(\rho\|E_A(\rho)) \leq e^{2gJ(1+2\beta|(A\partial)\partial|)}(\chi^0_{\min})^{-1}\,\mathrm{EP}_{\mathcal{L}^D_{A\partial}}(\rho), \tag{14}$$

where $\chi^0_{\min} > 0$ is the constant from (1).

This is not quite a bound on the cMLSI constant of the QMS generated by $\mathcal{L}_A$, because the entropy production on the RHS is the one of $\mathcal{L}_{A\partial}$ and not $\mathcal{L}_A$. However, this bound, together with the weak AT in Theorem 2.8 will suffice to obtain a suitable global MLSI constant. The idea behind the proof of this lemma (c.f. Appendix B.5) is to relate both the Davies conditional relative entropy and the entropy production of the Davies generators at inverse temperature β to the ones at 0. Then at 0 inverse temperature, the Davies semigroup is closely related to the depolarizing one allowing us to connect the two only by slightly enlarging the domain of the generator in the entropy production (c.f. Appendix B.6). A direct consequence of this lemma, Eq. (7) and a standard Grönwall argument is the following weak MLSI.

Corollary 2.13 (Quasi-polyweak MLSI, Theorem B.8). *In the context of Theorem 2.10, it holds that*

$$D(e^{t\mathcal{L}^D_\Lambda}(\rho)\|\sigma) \leq e^{-\alpha(\epsilon)t}D(\rho\|\sigma) + \epsilon \quad \textit{where} \quad \frac{1}{\alpha(\epsilon)} = \mathcal{O}\left(\exp\left(\operatorname{poly}\log\frac{N}{\epsilon}\right)\right).$$

The proof of Theorem 2.10 now combines the above quasi-polyweak MLSI Corollary 2.13 with the weak transport cost inequality of Theorem 2.11. For the detailed proof, see Theorem B.8.

2.5. Rapid Mixing Under Polynomial Local Gap

Having stronger assumptions on the gap of the local Davies generators, i.e. faster local mixing unsurprisingly yields tighter mixing bounds. The gap of the local Davies generator on $A \subset \Lambda$ is defined as the distance between the largest and second to largest eigenvalue of $\mathcal{L}_A^D$. A formal definition is given in (28). In [51,75], it is shown to lower bound the local cMLSI constant of $\mathcal{L}_A$ as $\alpha_c(\mathcal{L}_A) \geq \Omega\left(\frac{\lambda(\mathcal{L}_A)}{|A|}\right)$. Assuming a polynomial local gap with degree $\mu \in \mathbb{N}_0$ amounts to

$$\lambda(\mathcal{L}_A) \geq \Omega(|A|^{-\mu}) \quad \text{and hence} \quad \alpha_c(\mathcal{L}_A) \geq \Omega(|A|^{-1-\mu}) \quad \forall A \subset \Lambda, \tag{15}$$

which is equivalent to assuming a tightened version of Lemma 2.12 (or Lemma B.9). Note that technically an exact gap is not necessary and a result where the region of the generator is enlarged would suffice (analogous to Lemma 2.12), however, we will here and in the following always assume a gap for simplicity. With this assumption on the gap, we can improve the weak cMLSI as follows:

Corollary 2.14 (Polylogarithmic weak MLSI). *In the notation of Theorem 2.10 under the additional assumption of a local gap that is at most polynomially decaying with degree $\mu \in \mathbb{N}_0$, it holds that*

$$D(e^{t\mathcal{L}_\Lambda^D}(\rho)\|\sigma) \leq e^{-\alpha(\epsilon)t} D(\rho\|\sigma) + \epsilon, \tag{16}$$

where $\frac{1}{\alpha(\epsilon)} = \mathcal{O}\left(\left(\log \frac{N}{\epsilon}\right)^{D(1+\mu)}\right)$.

This weak MLSI combined with either Pinsker's inequality or Theorem 2.11 directly gives the following strengthened mixing time bounds.

Theorem 2.15 (Mixing under polygap). *Let $\{\mathcal{L}_{\Lambda_L}^D\}_L$ be a family of Davies Lindbladians corresponding to a family of (κ, r)-local, commuting, J-bounded Hamiltonians $\{H_{\Lambda_L}\}_L$ with uniform growth constant g. Assume that the local gap is at most polynomially decreasing with degree μ. Consider $\epsilon > 0$ and denote by $N = |\Lambda_L|$ the size of the lattice. Then, if the invariant state of $\mathcal{L}_{\Lambda_L}^D$, i.e. the Gibbs state σ^{Λ_L}, satisfies uniform exponential decay of its matrix-valued quantum conditional mutual information (MCMI) as defined in (2), the semigroup $\{e^{t\mathcal{L}_{\Lambda_L}^D}\}_{t\geq 0}$ satisfies*

$$t_{mix}^1(\epsilon) = \mathcal{O}\Big(\Big(\log \frac{N}{\epsilon^2}\Big)^{1+D(1+\mu)}\Big)_{\substack{N\to\infty\\ \epsilon\to 0}},$$

$$t_{mix}^{W_1}(\epsilon) = \mathcal{O}\Big(\Big(\log\Big(\frac{1}{\epsilon^2}\log\frac{N}{\epsilon^2}\Big)\Big)^{1+D(1+\mu)}\Big)_{\substack{N\to\infty\\ \epsilon\to 0}}.$$

For the non-asymptotic expressions for these mixing times, see Theorem C.2.

Polynomial assumptions on the local gap, have been used before to show rapid thermalization of Davies dynamics, e.g. for commuting quantum systems in 1D [11] under a condition equivalent to non-conditional MCMI decay. In [59], this was superseded, however, removing the required assumption on the local gap.

Remark 2. Note that instead of a uniform polynomial local gap, one can also require very high temperature, i.e. $\beta \sim \frac{1}{|\Lambda|}$ leading to a constant correction in Lemma 2.12, giving a result analogous to the one above only relying on the uniform decay of the MCMI.

To now get the main result Theorem 2.1, we can use any Lindblad simulation circuit to simulate $e^{t_{\mathrm{mix}}^{W_1}(\varepsilon)\mathcal{L}_\Lambda^D}$, which by the above, on any input state is close to σ in normalized W_1 distance. Using the one from [26, Theorem III.2] and our above mixing time bounds yields directly Theorem 2.1, though any Lindblad simulation circuit, for example, also [69] may be used. We call this algorithm *quasi-optimal* since it scales as $\mathcal{O}(N)$ up to quasi-logarithmic corrections, following the convention to denote sampling algorithms which scale as $\mathcal{O}(N)$ up to logarithmic corrections as "optimal". For more details, see Theorem C.3. A direct consequence of Theorems 2.10 and 2.15 is the following main result.

Corollary 2.16. *Let σ be a Gibbs state of a marginal commuting (see Appendix A.4.3), $(\kappa, r)-$local, $J-$bounded Hamiltonian on Λ_L at a uniformly suitably high temperature. Then the corresponding Davies semigroup has a W_1-mixing time bounded by*

$$t_{\mathrm{mix}}^{W_1}(\epsilon) = \text{quasi-poly}(\epsilon^{-1})_{\epsilon\to 0}\text{quasi-log}(N)_{N\to\infty} \tag{17}$$

and if additionally the gap of the local Davies generators is polynomial bounded, then

$$t_{mix}^{1}(\epsilon) = \text{poly}\log\left(N, \frac{1}{\epsilon}\right), \qquad t_{mix}^{W_1}(\epsilon) = \text{poly}\log\left(\log N, \frac{1}{\epsilon}\right). \tag{18}$$

Such systems include notably all quantum CSS codes and classical Gibbs states that satisfy complete analyticity [23,44,45]*, see Appendix A.4.2.*

We briefly comment on the relationship between the results in Theorems 2.10 and 2.15. The trace distance mixing time bound of Theorem 2.15 immediately yields a mixing time bound in normalized Wasserstein distance with the same parameter ε, while the reverse direction, i.e. from Theorem 2.10 to an estimate on the mixing time in trace distance, incurs a degradation from ε to $\frac{\varepsilon}{2N}$. More details can be found in Remark 7.

3. A Comparison with Quantum Gibbs Sampling

Here we further investigate the connections of the current submission with the vast recent literature in quantum Gibbs sampling.

In the presence of a local commuting Hamiltonian, the natural candidate way to design quantum Gibbs samplers is employing a Davies generator, which is local in this case. In [60], the authors showed that there is an equivalence between the existence of a positive spectral gap for the Lindbladian (which yields fast mixing) and a certain form of strong clustering in the Gibbs state of the Hamiltonian. This generalizes to the quantum setting the works [84,85].

However, this is not enough to yield rapid mixing. [10,11] showed the existence of a positive (logarithmically decaying with the system size) MLSI for translation-invariant 1D commuting local Hamiltonians at any positive temperature and thus rapid mixing. This result was improved to constant MLSI in [59], where rapid mixing was also shown for nearest neighbour Hamiltonians on hypercubic lattices at high-enough temperature, as well as for b-ary trees with small correlation length at high-enough temperature. This improves upon [33], where another efficient Gibbs sampler was provided for the former setting in terms of the so-called Schmidt generators, based on [21].

The very recent papers [48,57] present quantum generalizations of Glauber and Metropolis dynamics, inspired by the breakthrough [86]. Whereas [57] uses quantum phase estimation (QPE) in its algorithm, [48] is built on the Lindbladian introduced in [26,27], which can be regarded as a modification of Davies generator so that it is quasi-local for non-commuting Hamiltonians, in contrast to the usual Davies. A family of quantum Gibbs samplers satisfying the KMS detailed balance condition, which in particular includes the construction from [26], was presented in [39]. Moreover, the concurrent [20,76] show the existence of a positive spectral gap, and thus fast mixing, for the former Lindbladians at high-enough temperature, yielding efficiency for the Gibbs samplers. In [77], this result was further extended to establish rapid mixing under the existence of Lieb–Robinson bounds for the underlying Hamiltonian. All these works appeared after long-term efforts to implement Davies generators for non-commuting Hamiltonians in the most accurate way possible, as an exact implementation is impossible. Wocjan and Temme [89] dealt with this problem by assuming a rounding promise on the Hamiltonian, which was subsequently removed in [80] by using randomized rounding.

Additionally to all these approaches based on Davies (or similar) generators, there are some others based on Grover approaches [35,74], quantum imaginary time evolution [72] or quantum singular value transforms [52]. Some other approaches based on dissipation are [90], where the quantum Gibbs sampler is designed with simple local update rules, or [47], which is constructed from an energy functional inspired by the hypocoercivity of (classical) kinetic theory. Other works related to preparation of quantum Gibbs states are [46,54,64], among many others.

4. Outlook

The findings of this manuscript open up a series of natural questions in the contexts of entropy factorizations, mixing times, and decay of correlations on Gibbs states. As mentioned after Lemma 2.6, a strong entropy factorization is known to hold classically, see (6). Such a result presents an upper bound for the conditional relative entropy in ABC in terms of the sum of two conditional relative entropies in AB and BC, respectively, and a multiplicative error term, without the need for an additional additive error term. We strongly believe that something analogous should also hold in the quantum regime in quite some generality; however, a proof for this is still missing. Nevertheless, such a result is at least known to hold in the case that the second state is a tensor product [29]. Such a strong entropy factorization would directly lead to "strong" versions of the MLSI and TC inequality, with which one could directly conclude rapid mixing from the decay of MCMI by allowing decomposition of the lattice into local regions of constant size, following the argument of, for example, [33,59,71].

Refocusing on the main approach developed in this paper, based on the existence of a weak AT, the ingredients required to conclude rapid mixing are decay of the MCMI and a suitable local gap. One would hope that a polynomially bounded local gap, or more precisely an almost local gap (i.e. one can allow for constant size enlargements of the regions in the generator analogously to Lemma 2.12) should hold for the local Davies generators, assuming that the global Gibbs state satisfies some clustering condition, such as decay of the MCMI, or that it stems from a local CSS code. Proving this specifically for 2D Toric codes at every positive temperature, which we expect to hold, would imply an exponential improvement in the mixing time recently proved in [40], inspired in the original [2], and subsequently extended to 2D quantum double models in [63,67].

Finally, another relevant line of research is that of the MCMI decay for more general models, i.e. general (commuting) Hamiltonians, not only marginal commuting ones, which should be of independent interest, given the connections of the MCMI to the notions of conditional mutual information and mutual information, and their range of applicability. Furthermore, it would be interesting to see if the methods translate also to the non-commuting setting directly relating an information-theoretic measure—the MCMI—to the speed of the convergence of the samplers, e.g. the one in [26].

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A. Theoretical Framework

A.1. Basic Notation

We work with finite-dimensional Hilbert spaces, denoted by $\mathcal{H}$ or $\mathcal{K}$, and their associated algebras of bounded operators, $\mathcal{B}(\mathcal{H})$ and $\mathcal{B}(\mathcal{K})$, respectively. Throughout this discussion, we use several standard notations: the matrix trace is denoted by $\operatorname{tr}[\cdot]$, the Hilbert–Schmidt inner product by $\langle\cdot,\cdot\rangle$ and the Hilbert–Schmidt norm by $||\cdot||_2$. More generally, we use $||\cdot||_p$ for $p\in[1,\infty]$ to denote the Schatten-p-norms, with the operator norm given by $||\cdot||_\infty$. Operators are represented by capital Latin or lowercase Greek letters, depending on the context. For any operator $X\in\mathcal{B}(\mathcal{H})$, we write $X^\dagger$ for its adjoint and denote the subset of self-adjoint operators by $\mathcal{B}_{\mathrm{sa}}(\mathcal{H})$. The Hilbert–Schmidt adjoint of a linear map $\Phi:\mathcal{B}(\mathcal{H})\to\mathcal{B}(\mathcal{H})$ is written as Φ^*. We reserve $I_{\mathcal{H}}$ for the identity operator on $\mathcal{H}$ and id for the identity map on $\mathcal{B}(\mathcal{H})$, with $d_{\mathcal{H}}=\dim(\mathcal{H})$ denoting the dimension of $\mathcal{H}$. A quantum state (density operator) ρ is a positive semidefinite operator with unit trace, and we denote the set of states on $\mathcal{H}$ by $\mathcal{S}(\mathcal{H})$.

For any positive definite $X\in\mathcal{B}_{\mathrm{sa}}(\mathcal{H})$, we define two fundamental linear maps: the first is $\Gamma_X:\mathcal{B}(\mathcal{H})\to\mathcal{B}(\mathcal{H})$, given by $\Gamma_X(Y):=X^{\frac{1}{2}}YX^{\frac{1}{2}}$, and the second is the modular operator $\Delta_X:\mathcal{B}(\mathcal{H})\to\mathcal{B}(\mathcal{H})$, defined as $\Delta_X(Y)=XYX^{-1}$. In bipartite systems AB with $\mathcal{H}_{AB}=\mathcal{H}_A\otimes\mathcal{H}_B$, we use ρ_A to denote the reduced state $\operatorname{tr}_B[\rho_{AB}]$, where tr_B represents the partial trace over system B. By convention, we may write ρ_A as an operator in $\mathcal{B}(\mathcal{H}_{AB})$, to be understood as $\rho_A\otimes I_B$. The normalized partial trace $\mathbb{E}_B:\mathcal{B}(\mathcal{H}_{AB})\to\mathcal{B}(\mathcal{H}_{AB})$ is defined as $\mathbb{E}_B(X_{AB})=\operatorname{tr}_B[X_{AB}]\otimes I_B/d_B$.

For a bipartite state $\rho\in\mathcal{S}(\mathbb{C}^n\otimes\mathcal{H})$ and a quantum channel $\Psi:\mathcal{B}(\mathcal{H})\to\mathcal{B}(\mathcal{H})$, we write $\Psi(\rho):=(\mathrm{id}\otimes\Psi)(\rho)$, where a quantum channel is understood to be a linear completely positive trace-preserving map $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{K})$. The Umegaki relative entropy [87], a fundamental measure in quantum information theory, is defined for states ρ and σ as:

$$D(\rho\|\sigma) := \begin{cases} \operatorname{tr}[\rho \log \rho - \rho \log \sigma], & \text{if } \ker(\sigma) \subseteq \ker(\rho), \\ +\infty, & \text{otherwise,} \end{cases}$$

where $\ker(\cdot)$ denotes the kernel of an operator. By Pinsker's inequality, the relative entropy relates to the trace distance as:

$$||\rho - \sigma||_1 \leq \sqrt{2D(\rho\|\sigma)}\,. \tag{19}$$

For composite (or many-body) systems $\mathcal{H}_{\mathcal{I}} = \bigotimes_{i\in\mathcal{I}} \mathcal{H}_i$, we introduce the quantum Wasserstein distance of order 1 [41]. For two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_{\mathcal{I}})$, this distance is defined as the dual to a Lipschitz norm:

$$||\rho - \sigma||_{W_1} := \max_{H\in\mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}}), ||H||_L \leq 1} \langle H, \rho - \sigma \rangle\,, \tag{20}$$

where the Lipschitz norm is defined as

$$||H||_L := 2 \max_{i\in\mathcal{I}} \min_{\tilde{H}\in\mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}\setminus\{i\}})} \left|\left|H - I_i \otimes \tilde{H}\right|\right|_\infty \,, \tag{21}$$

for $H \in \mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}})$. The Wasserstein distance can also be regarded as a norm on the set of traceless self-adjoint operators and is an extensive quantity, scaling with the size of $\mathcal{I}$. This is reflected in the following relation with the trace distance, shown in [41, Proposition 5]: For any traceless $X \in \mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}})$ with $\operatorname{tr}_{\mathcal{J}}[X] = 0$, where $\mathcal{J} \subseteq \mathcal{I}$, we have

$$||X||_{W_1} \leq |\mathcal{J}|\, ||X||_1\,,. \tag{22}$$

For composite (or many-body) systems $\mathcal{H}_{\mathcal{I}} = \bigotimes_{i\in\mathcal{I}} \mathcal{H}_i$, we introduce the quantum Wasserstein distance of order 1 [41]. Informally, W_1 quantifies a minimal "local change": configurations that differ at a single site should be at unit distance, and differences that spread over many sites add up. This locality principle is enforced by a Lipschitz seminorm that measures how insensitive an observable can be made to each site by subtracting an operator acting on its complement.

For two quantum states $\rho, \sigma \in \mathcal{S}(\mathcal{H}_{\mathcal{I}})$, define

$$||\rho - \sigma||_{W_1} := \max_{H\in\mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}}),\, ||H||_L \leq 1} \langle H, \rho - \sigma \rangle\,,$$

where the Lipschitz norm is

$$||H||_L := 2 \max_{i\in\mathcal{I}} \min_{\tilde{H}\in\mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}\setminus\{i\}})} \left|\left|H - I_i \otimes \tilde{H}\right|\right|_\infty \,,$$

for $H \in \mathcal{B}_{\text{sa}}(\mathcal{H}_{\mathcal{I}})$. The Wasserstein distance can also be regarded as a norm on traceless self-adjoint operators and is extensive, scaling with $|\mathcal{I}|$.

Example 3. (reduction to the classical case [41]) If ρ and σ are diagonal in the computational basis, $||\rho-\sigma||_{W_1}$ coincides with the classical W_1 distance on $[\![1,d]\!]^{\mathcal{I}}$ equipped with the Hamming metric. In particular, for basis vectors $|x\rangle,|y\rangle\in[\![1,d]\!]^{\mathcal{I}}$ with Hamming distance

$$h(x,y)=\left|\{\, i\in\mathcal{I}: x_i\neq y_i \,\}\right|\,,$$

one has

$$\big\||x\rangle\langle x|-|y\rangle\langle y|\big\|_{W_1}=h(x,y)\,.$$

Hence, a single spin flip has unit cost,

$$W_1\Big(|0\rangle\langle 0|^{\otimes\mathcal{I}},\, |1\rangle\langle 1|\otimes|0\rangle\langle 0|^{\otimes(\mathcal{I}\setminus\{i\})}\Big)=1\,,$$

while flipping all sites costs $|\mathcal{I}|$,

$$W_1\Big(|0\rangle\langle 0|^{\otimes\mathcal{I}},\, |1\rangle\langle 1|^{\otimes\mathcal{I}}\Big)=|\mathcal{I}|\,.$$

This extensivity is reflected by the following relation with the trace norm [41, Proposition 5]: for any traceless $X\in\mathcal{B}_{\mathrm{sa}}(\mathcal{H}_{\mathcal{I}})$ with $\mathrm{tr}_{\mathcal{J}}[X]=0$ for some $\mathcal{J}\subseteq\mathcal{I}$,

$$\frac{1}{2}\,||X||_1\leq||X||_{W_1}\leq|\mathcal{J}|\,||X||_1\,. \tag{23}$$

Conditional expectations on von Neumann algebras play a central role in our analysis. For a von Neumann subalgebra $\mathcal{N}\subseteq\mathcal{B}(\mathcal{H})$, a conditional expectation onto $\mathcal{N}$ is a completely positive unital map $E^*_{\mathcal{N}}:\mathcal{B}(\mathcal{H})\to\mathcal{N}$ satisfying:

1. $E^*_{\mathcal{N}}(X)=X$ for all $X\in\mathcal{N}$;
2. $E^*_{\mathcal{N}}(VXW)=VE^*_{\mathcal{N}}(X)W$ for all $V,W\in\mathcal{N}$ and $X\in\mathcal{B}(\mathcal{H})$.

For such conditional expectations, the relative entropy exhibits special properties known as the chain rule and exact entropy factorization: given conditional expectation $E^*_{\mathcal{N}}:\mathcal{B}(\mathcal{H})\to\mathcal{N}$ and $E^*_{\mathcal{M}}:\mathcal{B}(\mathcal{H})\to\mathcal{B}(\mathcal{H})$ with $[E^*_{\mathcal{M}},E^*_{\mathcal{N}}]=0$, the chain rule states that for any state $\sigma=E_{\mathcal{N}}(\sigma)\in\mathcal{S}(\mathcal{H})$, and any state $\rho\in\mathcal{S}(\mathcal{H})$,

$$D(\rho\|\sigma)=D(\rho\|E_{\mathcal{N}}(\rho))+D(E_{\mathcal{N}}(\rho)\|\sigma)\,. \tag{24}$$

The exact entropy factorization on the other hand states that

$$D(\rho\|E_{\mathcal{N}}E_{\mathcal{M}}(\rho))\leq D(\rho\|E_{\mathcal{M}}(\rho))+D(\rho\|E_{\mathcal{N}}(\rho))\,. \tag{25}$$

For a full-rank state $\sigma>0$, we define a family of inner products, $s\in[0,1]$,

$$\langle X,Y\rangle_{s,\sigma}:=\langle X,\sigma^sY\sigma^{1-s}\rangle\,,$$

turning $\mathcal{B}(\mathcal{H})$ into a Hilbert space with norm

$$||X||^2_{1/s,\sigma}:=\langle X,X\rangle_{s,\sigma}\,.$$

In the cases $s=1/2$ and $s=1$ ($s=0$), $\langle X,Y\rangle_{s,\sigma}$ are also referred to as the KMS and GNS inner products, respectively. Note that for the KMS inner product, we will simply write $\langle\cdot,\cdot\rangle_\sigma$. Given the Hilbert space structure, we can define adjoints of linear maps $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$, calling a map Φ KMS- or GNS-symmetric if it is self-adjoint with respect to the respective inner product. Note that GNS symmetry implies KMS symmetry, as it implies commutativity with the modular operator, $[\Phi, \Delta_\sigma] = 0$. These inner products play a crucial role in the analysis of quantum Markov semigroups (QMS). A QMS $(\mathcal{P}_t)_{t\geq 0}$ is a semigroup of completely positive, trace-preserving maps on $\mathcal{B}(\mathcal{H})$ in Schrödinger and of completely positive, unital maps on $\mathcal{B}(\mathcal{H})$ in Heisenberg picture. Note that one is the dual of the other w.r.t. the Hilbert–Schmidt inner product. The unique generator of this semigroup, called a Lindbladian, is denoted by $\mathcal{L} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ and connected to the semigroup through the linear differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho = \mathcal{L}(\rho) \quad \text{with} \quad \rho(0) = \rho_0 \in \mathcal{S}(\mathcal{H})\,.$$

This motivates the notation $\mathcal{P}_t = e^{t\mathcal{L}}$, which we will use henceforth. A QMS is said to be KMS- or GNS-symmetric with respect to a full-rank state $\sigma \in \mathcal{S}(\mathcal{H})$ if the Hilbert–Schmidt dual of its generator $\mathcal{L}^* : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ exhibits the respective symmetry. For a generator of a QMS $\mathcal{L}^*$ which is further GNS-symmetric w.r.t $\sigma > 0$ the kernel $\ker(\mathcal{L}^*)$ is a von Neuman subalgebra with conditional expectation E^* given by $E^*[X] := \lim_{t\to\infty} e^{t\mathcal{L}^*}[X]$ for any $X \in \mathcal{B}(\mathcal{H})$.

A central question, and the primary focus of our work, concerns the speed of this convergence. Specifically, we investigate the convergence rate of a particular family of Lindbladians on quantum spin systems, measured in both the Wasserstein and trace distance.

A.2. Mixing, Rapid Mixing and Logarithmic Sobolev Inequalities for Open Quantum Systems

The study of convergence speeds for quantum Markov semigroups under different distance measures forms the core of this research. Here, we use the term "measure" informally as a way to compare states, rather than in its strict mathematical sense. While mixing phenomena are interesting even in simple systems, their study primarily focuses on composite systems. Therefore, we will work in the setting $\mathcal{H} = \bigotimes_{i\in\mathcal{I}} \mathcal{H}_i$. We will restrict ourselves to quantum Markov semigroups that are GNS-symmetric with respect to a full-rank state $\sigma > 0$, which guarantees both the convergence $e^{t\mathcal{L}} \to E$ as $t \to \infty$ and the von Neumann algebra structure of $\ker(\mathcal{L}^*)$. The mixing time in trace distance for $\varepsilon > 0$ is defined as

$$t^1_{\text{mix}}(\varepsilon) := \inf\{t \geq 0 : \forall \rho \in \mathcal{S}(\mathcal{H}), \left|\left|e^{t\mathcal{L}}(\rho) - E(\rho)\right|\right|_1 \leq \varepsilon\}\,. \tag{26}$$

Analogously, we define the mixing time in Wasserstein distance as

$$t^{W_1}_{\text{mix}}(\varepsilon) := \inf\{t \geq 0 : \forall \rho \in \mathcal{S}(\mathcal{H}), \left|\left|e^{t\mathcal{L}}(\rho) - E(\rho)\right|\right|_{W_1} \leq |\mathcal{I}|\varepsilon\}\,. \tag{27}$$

Our analysis aims to understand how these mixing times depend on both system size and ε. We classify systems as *mixing*, sometimes in literature also referred to as "fast mixing", when mixing times scale linearly with system size, while polylogarithmic scaling characterizes *rapidly mixing* or *rapid mixing* systems and quasi-logarithmic scaling characterized *quasi-rapid mixing*. Previous research has largely focused on trace distance, commonly analysing the generator's gap (c.f. [12,24,26,60])-the distance between the largest and second-largest eigenvalue of the Lindbladian. For KMS-symmetric Lindbladians, this gap is given by

$$\lambda(\mathcal{L}^*) := \inf_{X\in\mathcal{B}(\mathcal{H})} \frac{\langle X, -\mathcal{L}^*(X)\rangle_\sigma}{\left|\left|(\mathrm{id}-E^*)(X)\right|\right|_{2,\sigma}}\,. \tag{28}$$

By Grönwall's Lemma, we obtain $\left|\left|e^{t\mathcal{L}^*}(X)-E^*(X)\right|\right|_{2,\sigma} \le e^{-t\lambda(\mathcal{L}^*)} \left|\left|X-E^*(X)\right|\right|_{2,\sigma}$. Using Hölder's inequality on its variational form, this yields an estimate for the trace distance:

$$\begin{aligned}\left|\left|e^{t\mathcal{L}}(\rho)-\sigma\right|\right|_1 &\le \left|\left|\sigma^{-1/2}\right|\right|_\infty \sup_{X\in\mathcal{B}_{\mathrm{sa}}(\mathcal{H}_\mathcal{I}),\left|\left|X\right|\right|_\infty\le 1} \left|\left|e^{t\mathcal{L}^*}(X)-E^*(X)\right|\right|_{2,\sigma}\\ &\le 2\left|\left|\sigma^{-1/2}\right|\right|_\infty e^{-t\lambda(\mathcal{L}^*)}\,.\end{aligned}$$

Since $\left|\left|\sigma^{-1}\right|\right|_\infty$ typically scales exponentially with $|\mathcal{I}|$, the mixing times must be at least linear in $|\mathcal{I}|$ to compensate, assuming the gap is independent of $|\mathcal{I}|$. Alternative approaches have focused on analysing decay in relative entropy (c.f. [9,11,31,65]), attempting to establish (weak) modified logarithmic Sobolev inequalities (wMLSI). These inequalities relate the relative entropy to the entropy production of the semigroup's generator, defined as

$$\mathrm{EP}_\mathcal{L}(\rho) := -\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} D(e^{t\mathcal{L}}(\rho)\|\sigma) = -\mathrm{tr}[\mathcal{L}(\rho)(\log(\rho)-\log(\sigma))]\,. \tag{29}$$

The entropy production is always non-negative, following from monotonicity of the relative entropy under quantum channels. We say that $\mathcal{L}$ satisfies a (weak) modified logarithmic Sobolev inequality if there exist constants $c_1 > 0$ and $c_2 \ge 0$ such that for all $\rho \in \mathcal{S}(\mathcal{H})$

$$D(\rho\|E(\rho)) \le c_1\,\mathrm{EP}_\mathcal{L}(\rho) + c_2 \tag{wMLSI}$$

Applying Grönwall's Lemma yields

$$D(e^{t\mathcal{L}}(\rho)\|E(\rho)) \le e^{-t/c_1} D(\rho\|\sigma) + c_2\,,$$

and then by Pinsker's inequality (19), we obtain an estimate on the trace distance:

$$\left|\left|\rho-\sigma\right|\right|_1 \le \sqrt{2e^{-t/c_1}D(\rho\|\sigma)+2c_2},\,.$$

Since $D(\rho\|\sigma) \le \log\left|\left|\sigma^{-1}\right|\right|_\infty$, in the best case-assuming constant c_1 and $c_2 =$ 0-the mixing time need only be logarithmic in system size, classifying the system as rapid mixing. The novelty of our proofs lies in accepting positive c_2 which, however, decays with system size. As with many quantum–mechanical capacity quantities, there exists a notion of complete MLSI (cMLSI), defined as the supremum of MLSI of $\mathrm{id}_R \otimes \mathcal{L}$ over arbitrary reference systems R, denoted by $\alpha_c(\mathcal{L})$. This notation yields the relation

$$\alpha_c(\mathcal{L}_A\otimes \mathrm{id}_B + \mathrm{id}_A \otimes \mathcal{L}_B) \ge \min\{\alpha_c(\mathcal{L}_A), \alpha_c(\mathcal{L}_B)\}\,,$$

which was proved in [49] and is known for classical MLSI. Interestingly enough a relation between the gap of a GNS-symmetric QMS and the cMLSI was recently proved in [50,51,79] and we want to quote this estimate here as we later will use it in our analysis:

$$\alpha_c(\mathcal{L}) > \frac{\lambda(\mathcal{L})}{2\log(10C_{cb}(E))}\,, \tag{30}$$

where $C(E) := \inf\{c \geq 0 : \rho \leq cE(\rho),\ \forall \rho \in \mathcal{S}(\mathcal{H})\}$, $C_{cb}(E) := \sup_R C(\mathrm{id}_R \otimes E)$ is the (complete) Pimsner–Popa index, respectively.

A.3. Local Commuting Hamiltonian and Davies Generators on Lattices

In this paper, we consider D-dimensional hypercubes $\Lambda := \Lambda_L := [\![-L, L]\!]^D$ of side length $2\,L + 1$ consisting of $N = |\Lambda|$ sites, where each site hosts a qudit system $\mathcal{H}_i = \mathbb{C}^d$. The Hilbert space of the complete system is thus $\mathcal{H}_\Lambda = \bigotimes_{k\in\Lambda} \mathcal{H}_k$, and for any subsystem $A \subseteq \Lambda$, we have $\mathcal{H}_A = \bigotimes_{k\in A} \mathcal{H}_i$. In isolation, the system's dynamics are fully characterized by its Hamiltonian $H_\Lambda \in \mathcal{B}_{\mathrm{sa}}(\mathcal{H})$. When in thermal equilibrium with a heat bath at fixed inverse temperature $\beta > 0$, the system eventually assumes its thermal state, also known as the Gibbs state:

$$\sigma := \sigma^\Lambda := \frac{e^{-\beta H_\Lambda}}{\mathrm{tr}[e^{-\beta H_\Lambda}]}\,.$$

Our analysis focuses on a special class of Hamiltonians that are (κ, r)-local and commuting, with bounded interaction strength and connectivity. Specifically, there exists a representation

$$H_\Lambda = \sum_{A\subseteq\Lambda} h_A \otimes I_{\overline{A}}$$

where $\overline{A} := \Lambda \backslash A$ denotes the complement in Λ and $h_A \in \mathcal{B}_{\mathrm{sa}}(\mathcal{H}_A)$, satisfying the following conditions:

1. Commutativity: All terms $h_A \otimes I_{\overline{A}}$, $A \subseteq \Lambda$ pairwise commute;
2. (κ, r)-locality: $h_A = 0$ if either $|A| > \kappa$ or $\mathrm{diam}(A) > r$;
3. Bounded interaction strength: $\max_{A\subseteq\Lambda} ||h_A||_\infty =: J < \infty$;
4. Bounded connectivity: $\max_{k\in\Lambda} |\{A \subseteq \Lambda : h_A \neq 0, k \in A\}| =: g < \infty$.

Note that the *growth constant* g is a function of κ, r and D, and that κ, r are not independent. Despite that, we will treat them in this work as separate parameters.

When considering a family of Hamiltonians on increasing lattice sizes, such as those arising from an interaction on $\mathbb{Z}^D$, all the above constants are implicitly assumed to be independent of system size. In what follows, we will suppress the identity and simply write h_A, where the subindex indicates the support of the Hamiltonian term. For any set $R \subseteq \Lambda$, we define its closure dictated by the interaction structure of the Hamiltonian as

$$R\partial := \{k \in \Lambda : \exists A \subseteq \Lambda, h_A \neq 0, A \cap R \neq \emptyset, k \in A\}$$

and its boundary as $\partial R := R\partial\backslash R$. For our analysis, we introduce local Hamiltonians and local Gibbs states. The local Hamiltonian of $R \subseteq \Lambda$ is defined as

$$H_R := \sum_{A\subseteq R} h_A$$

and is strictly supported in R. Its corresponding local Gibbs state is $\sigma^R := \frac{e^{-\beta H_R}}{\mathrm{tr}[e^{-\beta H_R}]}$, which should not be confused with the marginal Gibbs state on R given by $\sigma_R := \mathrm{tr}_{\overline{R}}[\sigma]$. Using the locality and connectivity constraints, one readily obtains

$$||H_R||_\infty \leq gJ|R|\,, \tag{31}$$

which will prove useful in subsequent derivations.

We previously mentioned that, when a system is brought into contact with a heat bath, it is assumed to thermalize and eventually conform to the Gibbs state statistics. Assuming a specific interaction form between the bath and the system, i.e. a combined Hamiltonian of the form

$$H = H_\Lambda + H^{\mathrm{HB}} + \sum_{k\in\Lambda,\alpha} S_{\alpha,k} \otimes B_{\alpha,k}\,, \tag{32}$$

where H^{HB} is the bath Hamiltonian and $\{S_{\alpha,k} \otimes B_{\alpha,k}\}_{k\in\Lambda,\alpha}$ are self-adjoint single-site couplings one can derive an effective system dynamic through a weak coupling limit [81]. Note that we will not only assume that the $\{S_{\alpha,k}\}_\alpha$ are self-adjoint but for every k form a Kraus decomposition of the partial trace on the site k, i.e. $\sum\limits_\alpha S_{\alpha,k} \cdot S_{\alpha,k} = \mathrm{tr}_k[\,\cdot\,]$. An example would be the Pauli operators on $\mathbb{C}^2$. This choice ensures that in the trivial case $H_\Lambda = 0$ one recovers the depolarizing semigroup. The effective dynamics obtained by the weak coupling limits are governed by the so-called Davies Lindbladian

$$\mathcal{L}_\Lambda^{D,H}(\rho) := -i[H_\Lambda, \rho] + \mathcal{L}_\Lambda^D(\rho) \;= -i[H_\Lambda, \rho] + \sum_{k\in\Lambda} \mathcal{L}_k^D(\rho)\,, \tag{33}$$

for some local generators

$$\mathcal{L}_k^D(\rho) = \sum_{\omega,\alpha} \chi_{\alpha,k}^{\beta,\omega} \left(S_{\alpha,k}^\omega \rho S_{\alpha,k}^{\omega,\dagger} - \frac{1}{2}\left\{\rho, S_{\alpha,k}^{\omega,\dagger} S_{\alpha,k}^\omega\right\}\right). \tag{34}$$

The sum in (34) ranges over the index α of the local basis $\{S_{\alpha,k}\}$ as well as the Bohr frequencies ω of the Hamiltonian $H_{k\partial}$, i.e. all pairwise differences of its eigenvalues that arise as a result of the decomposition

$$e^{-itH_\Lambda} S_{\alpha,k} e^{itH_\Lambda} = e^{-itH_{k\partial}} S_{\alpha,k} e^{itH_{k\partial}} = \sum_\omega e^{it\omega} S_{\alpha,k}^\omega\,, \tag{35}$$

for arbitrary $t \in \mathbb{R}$. Note that the validity of the first equality relies on the commuting, local nature of the Hamiltonian, leading to the cancellation of all Hamiltonian terms that do not intersect k, thereby localizing the Bohr frequencies (i.e. their dependence only on $H_{k\partial}$). For better readability, we will not make the dependence on k explicit, however. Lastly the $\chi_{\alpha,k}^{\beta,\omega}$ in

(34) are the Fourier coefficients of the two-point correlation functions of the environment, which we assume to be uniformly bounded above and below: $0 < \chi^{\beta}_{\min} \leq \chi^{\beta,\omega}_{\alpha,k} \leq \chi^{\beta}_{\max}$. By assumptions made in the weak coupling limit, those coefficients satisfy the KMS condition, i.e. $\chi^{\beta,-\omega}_{\alpha,k} = e^{-\beta\omega}\,\chi^{\beta,\omega}_{\alpha,k}$ making the $\mathcal{L}^D_k$, GNS-symmetric w.r.t. to the Gibbs state of the same temperature. Similarly to the local Hamiltonians, we can define the Lindbladian on a set $R \subseteq \Lambda$ by $\mathcal{L}^D_R$ by restricting the sum in (33) to R. Unlike the local Hamiltonians, however, the local Lindbladians act non-trivially on $R\partial$ instead of R. One can define semigroups from these local Lindbladians that, due to their GNS symmetry also converge to projections, respectively, conditional expectation in the Heisenberg picture, we will denote by

$$E^*_A := \lim_{t\to\infty} e^{t\mathcal{L}^{D,*}_A} \quad \text{or equivalently} \quad E_A := \lim_{t\to\infty} e^{t\mathcal{L}^D_A}\,.$$

Note that these maps satisfy the following properties [11,15,60]:

1. $E_\emptyset = \mathrm{id}$ and $E_\Lambda = \mathrm{tr}[\,\cdot\,]\,\sigma^\Lambda$;
2. for each $A \subset A\partial \subseteq B \subseteq \Lambda$, σ^B is a fixed point of E_A;
3. for each $A \subseteq B \subseteq \Lambda$, $E_A E_B = E_B E_A = E_B$;
4. for any two regions $A, B \subseteq \Lambda$ such that $A\partial \cap B\partial = \emptyset$, $E_{A\sqcup B} = E_A E_B = E_B E_A$,

that will become useful later. Note that here and in the following we will use $\sqcup$ to denote a disjoint union of sets. By using the representation

$$E_A = \lim_{n\to\infty} \mathcal{R}^n_{\sigma,A}\,, \tag{36}$$

in terms of the Petz recovery

$$\mathcal{R}_{A,\sigma}(X) := \sigma^{1/2}\left(\sigma_{\overline{A}}^{-1/2}\mathrm{tr}_A[X]\sigma_{\overline{A}}^{-1/2}\right)\otimes I_A\,\sigma^{1/2}\,,$$

proved in [15, Theorem 1] and the locality of the Hamiltonian one readily verifies

$$E_A\mathbb{E}_A = E_A \quad \text{and} \quad \mathbb{E}_{A\partial}E_A = \mathbb{E}_{A\partial}\,. \tag{37}$$

Although $\mathcal{L}^D_A$ and E_A for $A \subseteq \Lambda$, as well as σ^A and σ_A, depend on the temperature, we will only make this dependence explicit where necessary to avoid a cluttering of indices.

A.4. Decay of Correlations in Gibbs States

The theory of thermalization in classical spin systems [71] suggests a strong connection between equilibrium and non-equilibrium properties of quantum spin systems. In classical systems, a well-established principle links the convergence rate of local stochastic dynamics to the decay of correlations in the Gibbs equilibrium measure. Specifically, the Glauber dynamics, which models the thermalization of a discrete spin system, rapidly approaches the equilibrium Gibbs measure if and only if correlations between spatially separated regions exhibit exponential decay with distance. Here, we consider a different notion of the decay of correlations and relate it to more standard classical notions.

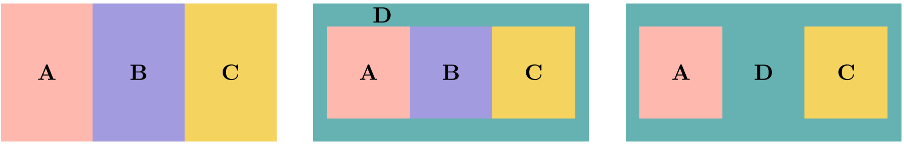

FIGURE 4. A lattice Λ is partitioned into four distinct regions, such that A and C are separated by either B or D or both. B is a system that is averaged over (through a partial trace) while D we condition on

A.4.1. Correlation Measures and the MCMI. The notion we want to introduce is that of the matrix-valued quantum conditional mutual information or MCMI for short. It has previously appeared as a tool to prove the decay of correlations in Gibbs states: more precisely, the decay of the (conditional) mutual information [62]. This result, unfortunately, contains a mistake, meaning we cannot recycle the decay of MCMI from there and have to rely on the toolbox of effective Hamiltonians, introduced later in Appendix A.4.3.

Definition A.1 (*Uniform exponential decay of matrix-valued quantum conditional mutual information*). Given any partition $\Lambda = A \sqcup B \sqcup C \sqcup D$ of the lattice, we define the *matrix-valued quantum conditional mutual information* (MCMI) of a state σ as

$$\mathbf{H}_\sigma(A : C|D) := \log \sigma_{ACD} + \log \sigma_D - \log \sigma_{AD} - \log \sigma_{CD} \,. \tag{38}$$

Then the Gibbs state $\sigma = \sigma^\Lambda$ is said to satisfy *uniform exponential decay of its matrix-valued quantum conditional mutual information* with constants K, ξ if there exist constants $K, \xi > 0$ independent of the regions A, B, C, D and such that

$$H_\sigma(A : C|D) := ||\mathbf{H}_\sigma(A : C|D)||_\infty \leq K|\Lambda|e^{-\frac{\mathrm{dist}(A,C)}{\xi}} \,. \tag{39}$$

Note that under slight abuse of notation, we will be referring to both $\mathbf{H}$ and H as the MCMI. To illustrate this decay and its geometric aspects, let us examine three two-dimensional situations in Fig. 4 which, despite their simplicity, capture all use cases later. The leftmost and the rightmost can be regarded as the extremal cases while the one in the middle in some sense interpolates the two. What all of them have in common and what is the crucial property is the separation of the regions A and C. This can be achieved either by averaging as in the case of B (operationally a partial trace) or through conditioning as in the case of D.

The system size-dependent prefactor we impose in our examples can be replaced by one that only depends on the minimum sizes of A and C (see Sect. D). For our purposes, the linear dependence is, however, sufficient. As the work in [62] suggests the MCMI is closely tied to other correlation measures. More precisely, given MCMI decay, one in particular has the decay of the conditional mutual information

$$I_\sigma(A : C|D) := -S(\sigma_{ACD}) - S(\sigma_D) + S(\sigma_{CD}) + S(\sigma_{AD})$$

$$= \mathrm{tr}[\sigma_{ACD}\, \mathbf{H}_\sigma(A:C|D)],$$

the mutual information

$$\begin{aligned} I_\sigma(A:C) &:= -S(\sigma_{AC}) + S(\sigma_A) + S(\sigma_C) \\ &= \mathrm{tr}[\sigma_{AC}\, \mathbf{H}_\sigma(A:C|\emptyset)], \end{aligned}$$

and in consequence also the decay of the covariance (see [13]). Note that $S(\sigma) := -\mathrm{tr}[\sigma \log \sigma]$ here denotes the von Neuman entropy. Besides its connection to quantum information-theoretic quantities, the MCMI also is connected to a notion of decay of correlation from classical statistical physics, namely complete analyticity. This comes as no big surprise since complete analyticity in classical systems implies a logarithmic Sobolev inequality and thereby also rapid mixing [23, Theorem 4.1].

A.4.2. MCMI Decay from Complete Analyticity in Classical Systems. Let us more closely investigate the connection to the *complete analyticity* also called the Dobrushin–Shlosman condition of classical spin systems [23,44,45]. We will show that indeed complete analyticity implies uniform decay of the MCMI. Although we believe that also the reverse implication holds, we are still missing the proof as of the day of writing.

Proposition A.2. *Given a classical spin lattice system with interactions that satisfy complete analyticity as in* [23]*, then it also satisfies uniform decay of the MCMI, i.e. there exist* $K, \xi > 0$ *s.t.*

$$\|\boldsymbol{H}_\sigma(A:C|D)\|_\infty = \left\|\log \frac{\sigma_{ACD}\sigma_D}{\sigma_{AD}\sigma_{CD}}\right\|_\infty = \left\|\log \frac{\sigma_{A|CD}}{\sigma_{A|D}}\right\|_\infty \leq K|C|e^{-\frac{\mathrm{dist}(A,C)}{\xi}} \tag{40}$$

Note that in this section we will follow the notation of [71] mostly (re)defining the necessary objects ad hoc. We will further suppress all temperature dependence to improve readability.

Proof. Let us consider the single spin system $S = \{-1, 1\}$ with counting measure leading to a full state space $\Omega = \{\omega : \mathbb{Z}^D \to S\}$. We will write $\omega_V = \omega|_V$ for the restriction of $\omega \in \Omega$ to V. The interaction which completely characterizes the model is given as $J : \{V \Subset \mathbb{Z}^D\} \to \mathbb{R}$ and we write J_V instead of $J(V)$. Here and in the following $V \Subset \mathbb{Z}^D$ refers to finite subsets. The corresponding Hamiltonian on $V \Subset \mathbb{Z}^D$ is given by

$$H_V(\omega) = -\sum_{A:\, A\cap V \neq \emptyset} J_A \prod_{x\in A} \omega(x)\,,$$

and the Gibbs measure for the boundary condition $\tau \in \Omega$ as

$$\mu_V^\tau(\omega) = \begin{cases} (Z_V^\tau)^{-1} \exp[-H_V(\omega)] & \text{if } \omega_{\overline{V}} = \tau_{\overline{V}}\,, \\ 0 & \text{else.} \end{cases}$$

Here and in the rest of this section, a line over a set denotes the complement w.r.t. $\mathbb{Z}^D$, e.g. $\overline{V} = \mathbb{Z}^D \backslash V$ in the above definition. Interestingly for $V \subset \Lambda \Subset$

$\mathbb{Z}^D$, we find that μ_V^τ alternatively can be written as

$$\mu_V^\tau(\omega) = \begin{cases} (\tilde{Z}_V^\tau)^{-1} \exp[-H_\Lambda(\omega)] & \text{if } \omega_{\overline{V}} = \tau_{\overline{V}}\,, \\ 0 & \text{else.} \end{cases} \tag{41}$$

That is, the Hamiltonian on V can be replaced by the Hamiltonian on Λ through a suitable change of normalization $Z_V^\tau \to \tilde{Z}_V^\tau$. This is due to the conditioning with τ which as a consequence ensures the existence of a $c > 0$ only dependent on $\tau_{\overline{V}}$, s.t. for all σ_V, $H_V(\sigma_V\tau_{\overline{V}}) + c = H_\Lambda(\sigma_V\tau_{\overline{V}})$. The marginal of a Gibbs distribution on $\Delta \subset V$ is defined as $\mu_{V,\Delta}^\tau : \Omega \to \mathbb{R}$ with

$$\mu_{V,\Delta}^\tau(\omega) = \sum_{\nu:\nu|_\Delta = \omega|_\Delta} \mu_V^\tau(\nu)\,. \tag{42}$$

By the definition of [23], a system is *completely analytic* if there exists $K > 0$, $\xi > 0$ such that for all $V \Subset \mathbb{Z}^D$, $x \in \partial V := \{x \in \overline{V} :\ \mathrm{dist}(x, V) \leq r\}$, $\Delta \subset V$ and $\tau, \tau' \in \Omega$ with $\tau(y) = \tau'(y)$ for all $y \neq x$

$$\left\| \frac{\mu_{V,\Delta}^\tau}{\mu_{V,\Delta}^{\tau'}} - 1 \right\|_\infty \leq K e^{-\frac{\mathrm{dist}(x,\Delta)}{\xi}}\,. \tag{43}$$

Now let us try to make the connection to Definition A.1. Let us first fix the equivalent of σ^Λ of $\Lambda \Subset \mathbb{Z}^D$. Since quantumly we are considering only Hamiltonians on finite lattice Λ let us fix a boundary condition $\nu \in \Omega$ and set classical analogue of our Gibbs measure σ_Λ on Λ to be $\mu^\nu = \mu_\Lambda^\nu : \Omega \to \mathbb{R}$. Marginals of this distribution are defined through (42) and for $\Sigma \subset \Lambda$ denoted by μ_Σ^ν. We further adapt the shorthand notation $\mu_{\Sigma|\Lambda}^\nu := \frac{\mu_{\Sigma\Lambda}^\nu}{\mu_\Lambda^\nu}$. Incorporating the implicit boundary condition explicitly, Definition A.1 states: There exist $\xi > 0$ and $K > 0$ such that for all partitions $\Lambda = A \sqcup B \sqcup C \sqcup D$

$$\left\| \log \frac{\mu_{A|CD}^\nu}{\mu_{A|D}^\nu} \right\|_\infty \leq K|\Lambda| e^{-\frac{\mathrm{dist}(A,C)}{\xi}}\,. \tag{44}$$

First note that it suffices to investigate the case when C only contains a single site, as we can write

$$\left\| \log \frac{\mu_{A|CD}^\nu}{\mu_{A|D}^\nu} \right\|_\infty \leq \sum_{i=1}^{|C|} \left\| \log \frac{\mu_{A|\{x_i\}D_i}^\nu}{\mu_{A|D_i}^\nu} \right\|_\infty \tag{45}$$

where $D_1 = D$, $D_i = D_{i-1} \sqcup C_{i-1}$ and $C_i = \{x_i\}$ with $\{x_i\}_{i=1}^{|C|}$ an enumeration of C. In the following, we will drop ν for better readability. By the above argument, we are now in the setting of $|C| = 1$. Let us assume $\tau \in \Omega$ to be the optimizer such that $\left\| \log \frac{\mu_{A|CD}}{\mu_{A|D}} \right\|_\infty = \log \frac{\mu_{A|D}(\tau)}{\mu_{A|CD}(\tau)}$. In the case that the optimum is achieved for the reversed fraction, the convexity of $x \mapsto \frac{1}{x}$ gives an inequality instead of equality in the second line of (46) while also the roles of τ and τ' change, however, leaving the rest of the argument invariant. Let us denote by τ' the element of Ω which agrees with τ on $\overline{C}$ but has a flipped spin at site C. By $\log(x+1) \le x$, the fact that $\mu_{C|D}(\tau) = 1 - \mu_{C|D}(\tau')$ with both $\mu_{C|D}(\tau)$ and $\mu_{C|D}(\tau')$ in $[0,1]$, we get

$$\begin{aligned}\log \frac{\mu_{A|D}(\tau)}{\mu_{A|CD}(\tau)} &\le \frac{\mu_{A|D}(\tau)}{\mu_{A|CD}(\tau)} - 1 = \frac{\mu_{A|CD}(\tau)\mu_{C|D}(\tau) + \mu_{A|CD}(\tau')\mu_{C|D}(\tau')}{\mu_{A|CD}(\tau)} - 1 \\ &= \mu_{C|D}(\tau')\left(\frac{\mu_{A|CD}(\tau')}{\mu_{A|CD}(\tau)} - 1\right) \le \left|\frac{\mu_{A|CD}(\tau')}{\mu_{A|CD}(\tau)} - 1\right| = \left|\frac{\mu^{\tau'}_{AB,A}(\tau)}{\mu^{\tau}_{AB,A}(\tau)} - 1\right| \\ &\le \left\|\frac{\mu^{\tau'}_{AB,A}}{\mu^{\tau}_{AB,A}} - 1\right\|_\infty . \end{aligned} \tag{46}$$

In the last equality, we used the fact that

$$\mu^{\tau}_{AB,A}(\omega) = \mu_{A|CD}(\omega_A \tau_{\overline{A}}) \, , \tag{47}$$

which one readily obtains from (41). As the system satisfies complete analyticity, this last norm is uniformly bounded by $K e^{-\frac{\mathrm{dist}(A,C)}{\xi}}$ which for the general setting of (45) allows us to conclude

$$\left\|\log \frac{\mu_{A|CD}}{\mu_{A|D}}\right\|_\infty \le K|C| e^{-\frac{\mathrm{dist}(A,C)}{\xi}} .$$

□

A.4.3. The Effective Hamiltonians as a Tool to Prove Correlation Decay. Let us conclude the section about the theoretical framework by introducing a tool which has proved useful when showing the decay of some correlation measures for explicit models. This tool is called effective Hamiltonians and has appeared in [62] and was further developed in [14]. We will restrict our discussion to the case where $\Lambda \subseteq \mathbb{Z}^D$ is a finite hypercube and address the Hamiltonian directly, whereas the results in [14] apply to interactions on general graphs. Next, we introduce an essential norm originally developed for interactions; here, we define it specifically for Hamiltonians with a fixed local representation (cf. (31)): For H_Λ with representation as in (31) and $\mu > 0$ the interaction norm is defined as

$$|||H_\Lambda|||_\mu := \sup_{x\in\Lambda} \sum_{X\subseteq\Lambda : x\in X} ||h_X||_\infty \, e^{\mu\, \mathrm{diam}(X)} . \tag{48}$$

Slightly adapting the definition from [14, Definition 3.1], we say that H_Λ admits a *strong local effective Hamiltonian at inverse temperature* $\beta > 0$ if for every $A \subseteq \Lambda$ there exists a Hamiltonian with a local decomposition $\tilde{H}^A = \sum_{X\subseteq\Lambda} \tilde{h}^A_X$ such that

1. $\tilde{h}^A_X$ is supported in $X \cap A$ for every $X \subseteq \Lambda$;
2. For $A' \subseteq \Lambda$, $\tilde{h}^A_X = \tilde{h}^{A'}_X$ for all $X \subseteq \Lambda$ for which $X \cap A' = X \cap A$;
3. One has $\log d^{-1}_{\overline{A}} I_{\overline{A}} \otimes \mathrm{tr}_{\overline{A}}[e^{-\beta H_\Lambda}] = \sum_{X\subseteq\Lambda} \tilde{h}^A_X = \tilde{H}^A$.

The existence of an effective Hamiltonian alone does not suffice to ensure the decay of the MCMI, as the above definition lacks information on the system's locality. In general, this decay will require an additional locality assumption (see Appendix D.1). However, when the system Hamiltonian is also marginal commuting, [14, Theorem 4] directly guarantees the existence of an effective

Hamiltonian with explicit decay properties. This result allows us to establish uniform decay of the MCMI at high temperatures in Appendix D.2. To clarify, we define what it means for H_Λ to have commuting marginals: We say that $H_\Lambda = \sum_{A\subseteq\Lambda} h_A$ is *marginal commuting* if there exists a commuting algebra $\mathcal{A}$ generated by all the local terms h_A closed under the application of $\mathbb{E}_A$ for any subset $A \subseteq \Lambda$, i.e. $\mathbb{E}_A[\mathcal{A}] \subset \mathcal{A}$. This definition is exactly [14, Definition 3.5] and is named here to highlight that, under this assumption, arbitrary marginals of the Gibbs state commute.

B. Proof Techniques—Extended

B.1. A General Weak Entropy Factorization

This section is dedicated to the proof and a short discussion of a weak entropy factorization. More precisely, a splitting of the conditional relative entropy defined for $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\overline{A}}$ with $A \subseteq \Lambda$ as

$$D_A(\rho\|\sigma) := D(\rho\|\sigma) - D(\rho_{\overline{A}}\|\sigma_{\overline{A}}) \tag{49}$$

for $\rho, \sigma \in \mathcal{S}(\mathcal{H}_\Lambda)$. We can understand this as the relative entropy in the A subsystem conditioned on the rest of the lattice. It made its first appearance in [28] where it also was connected to the Davies channel on A, more precisely, the following important inequality was shown:

$$D_A(\rho\|\sigma) \leq D(\rho\|E_A(\rho))\,. \tag{50}$$

Naturally $D_\Lambda(\rho\|\sigma) = D(\rho\|E_\Lambda(\sigma)) = D(\rho\|\sigma)$.

The following lemma details the weak entropy factorization which is one of the key ingredients in our main results.

Lemma B.1 (Weak entropy factorization). *Let $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C \otimes \mathcal{H}_D$ and $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ with $\sigma > 0$, then*

$$D_{ABC}(\rho\|\sigma) \leq D_{AB}(\rho\|\sigma) + D_{BC}(\rho\|\sigma) + ||\mathbf{H}_\sigma(A : C|D)||_\infty\,. \tag{51}$$

Proof. We have that

$$\begin{aligned}
&D_{ABC}(\rho\|\sigma) - D_{AB}(\rho\|\sigma) - D_{BC}(\rho\|\sigma) && \text{(DPI)}\\
&\quad = -D(\rho_D\|\sigma_D) - D(\rho_{ABCD}\|\sigma_{ABCD}) + D(\rho_{CD}\|\sigma_{CD}) + D(\rho_{AD}\|\sigma_{AD})\\
&\quad \leq -D(\rho_D\|\sigma_D) - D(\rho_{ACD}\|\sigma_{ACD}) + D(\rho_{CD}\|\sigma_{CD}) + D(\rho_{AD}\|\sigma_{AD})\\
&\qquad\qquad \leq \operatorname{tr}[\rho_{ACD}(\log\sigma_{ACD} + \log\sigma_D - \log\sigma_{AD} - \log\sigma_{CD})] && \text{(SSA)}\\
&\qquad\qquad \leq ||\mathbf{H}_\sigma(A : C|D)||_\infty\,. && \text{(Hölder)}
\end{aligned}$$

□

The correction's additive nature is why we only manage to obtain the weak approximate tensorization for the Davies (Appendix B.3) and as a consequence a weak modified logarithmic Sobolev inequality and a weak transport cost inequality (Appendix B.4). Lifting the above inequality to one which has a multiplicative correction would strengthen all results in this paper to their strong counterparts and further allow one to eliminate the requirement for a polynomial gap in Appendix C.2 and in the strengthening of the W_1 mixing in Appendix C.2. At the date of writing, we only managed to prove the inequality in the fully classical case where we obtain

$$(1 - ||\exp(\mathbf{H}_\sigma(A:C|D)) - I||_\infty) D_{ABC}(\rho\|\sigma) \leq D_{AB}(\rho\|\sigma) + D_{BC}(\rho\|\sigma)\,.$$

Note that in the special case where D is empty, i.e. $\dim \mathcal{H}_D = 1$, the authors in [29] show a multiplicative correction. Their proof yields a slightly altered factor, classically equivalent to the above bound.

B.2. A Coarse-Graining of the Hypercubic Lattice

In this section, we present the construction of the coarse-graining w.r.t. which we apply the weak entropy factorization (c.f. Appendix B.1) to prove the weak approximate tensorization later (c.f. Appendix B.3). This coarse-graining will be at the heart of its proof and our divide-and-conquer scheme involving both exact and weak approximate tensorizations at every step.

We do this by recalling the construction of a coarse-graining of the two-dimensional hypercube due to [19] and extending it to any dimension $D \in \mathbb{N}$. We denote by $\Lambda_L := [\![-L, L]\!]^D \subset \mathbb{Z}^D$ the D-dimensional hypercube of side length $2\,L+1$. We denote by $\Lambda_L := [-L, L]^D \cap \mathbb{Z}^D$ the lattice points in the D-dimensional hypercube of side length $2L+1$. Our decomposition relies on the fixation of three parameters:

1. $c \in \mathbb{N}$ with $c \geq r$ the *overlap length*, ensuring a sufficiently fast decay when using a weak approximate tensorization to consecutively cut out cells reducing the extensive dimension in every step.
2. $k \in \mathbb{N}$ with $k \geq r$ the *buffer length*, separating cells of the same dimensionality such that the corresponding conditional expectations commute and hence allow us to use tensorization of the relative entropy.
3. $\ell \in \mathbb{N}$ with $L \geq \ell$ and ℓ odd the "extensive" side length of the cells we decompose into. As we will consecutively reduce the "extensive" dimensions until we reach zero subtracting ℓ in every step by $2(k+c)$, we further require $\ell > 2D(k+c)$.

Remark 4. In the first step of the decomposition, we aim to divide Λ_L into hypercubes of side length bounded by ℓ. This might not be possible due to the choice of ℓ we made prior as divisibility of $2L+1$ by ℓ may not be guaranteed. We can, however, always embed Λ_L into some Λ_{L+r} with $r \in [\![1, l-1]\!]$ such that $2(L+r)+1$ is now divisible by ℓ, when ℓ is odd. Since we are interested in how certain quantities scale in ℓ we may always choose ℓ odd, however, we will overlook this subtlety.

The original Hamiltonian is likewise embedded by padding it with zeros, leading to a Gibbs state of the form $\sigma^{\Lambda_L} \otimes \pi_{\text{rest}} \in \mathcal{S}(\mathcal{H}_{\Lambda_{L+r}})$ where π_{rest} is the maximally mixed state on $\Lambda_{L+r} \backslash \Lambda_L$. The (κ, r)-locality and interaction strength of both Hamiltonians, original and padded, agree, due to the specific embedding. Furthermore, the condition of the global and local gaps and the decay of MCMI hold for the embedded if and only if they hold for the original

Hamiltonian. This means we employ our proof for the embedded version and immediately get a result for the original one.

Consider the example of the MLSI, where we denote by $\mathcal{L}$ the Davies Lindbladian in Λ_L and by $\mathcal{L}_{\text{rest}}$ that of $\Lambda_{L+r} \setminus \Lambda_L$. The separation between them is due to the embedding which in addition gives $[\mathcal{L}, \mathcal{L}_{\text{rest}}] = 0$. Assuming the embedded system has MLSI, we have $D(e^{t(\mathcal{L}+\mathcal{L}_{\text{rest}})}(\rho')\|\sigma \otimes \pi_{\text{rest}}) \leq e^{-\frac{t}{c_1}} D(\rho'\|\sigma\otimes\pi_{\text{rest}})$. By setting $\rho' = \rho\otimes\pi_{\text{rest}}$ and utilizing commutativity of the Lindbladians, we obtain $D(e^{t\mathcal{L}}(\rho)\otimes\pi_{\text{rest}}\|\sigma\otimes\pi_{\text{rest}}) \leq e^{-\frac{t}{c_1}} D(\rho\otimes\pi_{\text{rest}}\|\sigma\otimes\pi_{\text{rest}})$. Through the tensorization of relative entropy, we can conclude that c_1 also serves as an MLSI constant for the original system. This approach extends to other contexts (e.g. the decay of Wasserstein distance). We have established that all conditions imposed on the original system remain unchanged for the embedding, meaning the only change in the obtained bounds is in the system size. Notably, $|\Lambda_{L+r}| \leq 2^D|\Lambda_L|$ since $\ell \leq L$, meaning the correction only depends on the lattice dimension and does not affect the scaling with system size. Throughout the paper, Λ_L is treated as general. However, in the proofs, we always assume without loss of generality that ℓ divides $(2L + 1)$. When providing explicit constants for the bounds, we include the correction factor 2^D.

In the process of constructing the coarse-graining, we will define various cells of dimension a, where a ranges from 0 to $D - 1$. These cells should be understood as embedded within the corresponding D-dimensional hypercube for which $e_1, \dots, e_D$ denote the canonical orthonormal basis. We begin by partitioning the D-dimensional hypercube $C^D := \Lambda_L$ into smaller D-hypercubes, each with side length ℓ. This can for example be done by choosing the anchor point of the cell from Λ_L (these are the points that differ from one another along the main coordinate directions by multiples of ℓ), i.e. $x_i \in \Lambda_L$ and setting

$$C^{\partial}_{D,i} = \left(\times_{j=1}^{D}[x_{i,j} - 1/2, x_{i,j} + l - 1/2)\right) \cap \Lambda_L\,. \tag{52}$$

We denote these partition elements as $\{C^{\partial}_{D,i}\}_{i\in\mathcal{I}_D}$ with $\mathcal{I}_D = [\![1, ((2\,L + 1)/\ell)^D]\!]$. For each cell $C^{\partial}_{D,i}$, we define two subsets:

1. The interior of $C^{\partial}_{D,i}$, excluding a boundary layer of buffer length k:
$$C_{D,i} := \{v \in C^{\partial}_{D,i} :\ \operatorname{dist}(v, \overline{C^{\partial}_{D,i}}) > k\}\,;$$
2. A further restricted interior, excluding a boundary layer of width $k + c$, i.e excluding buffer and overlap length:
$$\mathring{C}_{D,i} := \left\{v \in C^{\partial}_{D,i} :\ \operatorname{dist}(v, \overline{C^{\partial}_{D,i}}) > k + c\right\}\,.$$

We define the aggregate sets:

$$C_D := \bigsqcup_{i\in\mathcal{I}_D} C_{D,i} \qquad \mathring{C}_D := \bigsqcup_{i\in\mathcal{I}_D} \mathring{C}_{D,i},$$

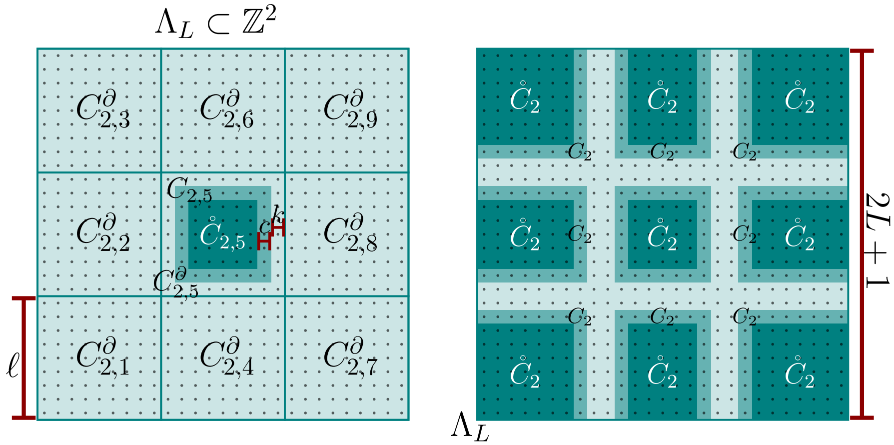

FIGURE 5. Splitting of $\Lambda_L = C^D$ for $D = 2$ as described in the text. On the left side, Λ_L has been split into 9 2-cells. For the cell in the middle $C_{2,5}$ and $\mathring{C}_{2,5}$ are explicitly represented. On the right side, C_2 is the union of the midblue and dark blue regions, and $\mathring{C}_2$ is the union of the dark blue regions (Color figure online)

where $\bigsqcup$ denotes the disjoint union. Finally, we set $C^{D-1} := \Lambda_L \setminus \mathring{C}_D$. For a visual representation of this construction in two dimensions ($D = 2$), refer to Fig. 5.

We continue the deconstruction by defining cells of lower extensive dimensions. For each dimension $a \in [\![1, D-1]\!]$, we iteratively construct a-cells from C^a as follows: For $a \in [\![1, D-1]\!]$, let $C^a \subseteq \Lambda_L$ be the union of all remaining vertices after having peeled off $\bigsqcup_{b=a+1}^{D} \mathring{C}_b$. Define

$$C^\partial_a := \{\, v \in C^a : \exists\, j \in \mathbb{Z},\ e_b \in \{e_1, \ldots, e_D\} \text{ with } v + j\, e_b \in \overline{C^a} \,\},$$

i.e. the connecting skeleton between neighbouring sets $\mathring{C}_{a+1,i}$, $\mathring{C}_{a+1,k}$, respectively.

1. By construction, C^∂_a decomposes into maximal connected components with respect to the nearest neighbour graph, where vertices x and y are connected if their Euclidean distance satisfies $\operatorname{dist}(x, y) \le 1$. We denote these components by $\{C^\partial_{a,i}\}_{i\in\mathcal{I}_a}$, giving us the disjoint union

$$C^\partial_a = \bigsqcup_{i\in\mathcal{I}_a} C^\partial_{a,i}.$$

Each component $C^\partial_{a,i}$ is a hyperrectangle that contains precisely those vertices of C^a lying within distance $2(D - a + 1)(k + c)$ of the same $(a-1)$-face. Equivalently, C^∂_a can be viewed as a disjoint union of fattened a-cells. This follows from our construction: each component $C^\partial_{a,i}$ forms a hyperrectangle that bridges two neighbouring hyperrectangles $\mathring{C}_{a+1,i}$

and $\mathring{C}_{a+1,k}$. Since these hyperrectangles had their sides aligned during the previous construction step, the resulting components maintain this rectangular structure. An illustration of this decomposition is provided in Fig. 6.

2. First, we identify the set of vertices that "join" each pair of neighbouring $\mathring{C}_{a+1,i}$:
$$C_a^\partial := \left\{v \in C^a :\ \exists j \in \mathbb{Z}, b \in [\![1, D]\!] : v + je_b \in \overline{C^a}\right\} .$$
3. By construction, C_a^∂ is a disjoint union of fattened a-cells, which we denote as $C_{a,i}^\partial$ with index set $\mathcal{I}_a$.
4. We again define the two subsets:
$$C_{a,i} := \{v \in C_{a,i}^\partial :\ \mathrm{dist}(v, \overline{C_{a,i}^\partial} \cap C^a) > k\} ,$$
$$\mathring{C}_{a,i} := \{v \in C_{a,i}^\partial :\ \mathrm{dist}(v, \overline{C_{a,i}^\partial} \cap C^a) > k + c\}$$
separated from the skeleton that remains after removal of $\bigsqcup_{b=a+1}^D \mathring{C}_b$ by buffer and overlap plus buffer length, respectively. We set the aggregated sets to be
$$C_a := \bigsqcup_{i\in\mathcal{I}_a} C_{a,i}, \qquad \mathring{C}_a := \bigsqcup_{i\in\mathcal{I}_a} \mathring{C}_{a,i} .$$
5. Finally, we define the set for the next lower dimension: $C^{a-1} := C^a \setminus \mathring{C}_a$.

This process is iterated for decreasing values of a, creating a hierarchical structure of cells of different extensive dimensions. For a visual representation of this construction for $a = 1$ in two dimensions ($D = 2$), refer to Fig. 6.

Lastly for $a = 0$ we set $C^0 := C^1 \setminus \mathring{C}_1$ and $\mathring{C}_{0,i} := C_{0,i} := C_{0,i}^\partial$ so that $C^0 = \bigsqcup_{i\in\mathcal{I}_0} C_{0,i}$. This is shown on the left side of Fig. 7. Let us summarize the properties of the above construction in the following lemma.

Lemma B.2. *The decomposition of Λ_L described in the previous paragraphs satisfies the following properties:*

1. *For each $a \in [\![0, D]\!]$, $\mathring{C}_a := \bigsqcup_{i\in\mathcal{I}_a} \mathring{C}_{a,i}, C_a := \bigsqcup_{i\in\mathcal{I}_a} C_{a,i}$, and $C_a^\partial := \bigsqcup_{i\in\mathcal{I}_a} C_{a,i}^\partial$ are unions of disjoint sets, with size bounded as $|\mathring{C}_{a,i}| \leq |C_{a,i}| \leq |C_{a,i}^\partial| \leq \ell^D$.*[2]
2. *$C^0 = C_0 = \mathring{C}_0 = \bigsqcup_{i\in\mathcal{I}_0} C_{0,i}^\partial$ is a disjoint union of "fat" 0-cells, in the shape of D-dimensional "crosses", with each $C_{0,i}^\partial$ included in a hypercube of side length $2D(k + c)$ and distance*
$$\mathrm{dist}(C_{0,i}^\partial, C_{0,j}^\partial) \geq \ell - 2D(k + c) > 0$$
from each other.
3. *The hierarchy $\{C_a\}_{a=0}^D$ induces a suitably overlapping coarse-graining of Λ_L, i.e. $\bigcup_{a=0}^D C_a = C^D = \Lambda_L$, and each site $x \in \Lambda_L$ is included in at most $D + 1$ sets $\{C_{a,i}\}_{a,i}$.*

[2] This is a non-tight bound and it holds that $|C_{a,i}| \leq [2(D - a)(k + c)]^{D-a}[\ell - 2(D - a)(k + c)]^a \leq \ell^D$.

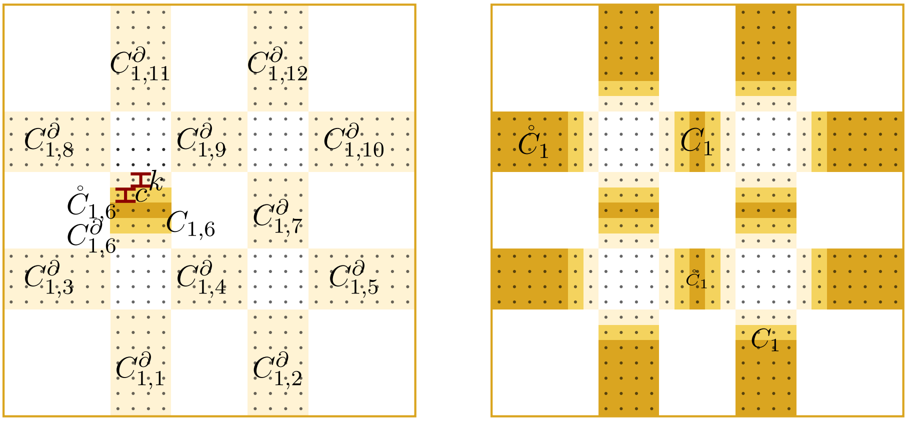

Figure 6. The figure shows the construction of C_1^{∂}, contained in $C^1 = \Lambda_L \backslash \mathring{C}_1$. On the left side, C^1 is the whole region, and C_1^{∂} has been split into several fattened 1-cells, respectively; we represent explicitly, $C_{1,6}$, $\mathring{C}_{1,6}$. On the right side, $\mathring{C}_1$ is the union of the dark green regions, $C_1 \backslash \mathring{C}_1$ is that of the medium green ones, and $C_1^{\partial} \backslash C_1$ is the union of the lighter green, dashed regions without the light green squares in the centre, however (Color figure online)

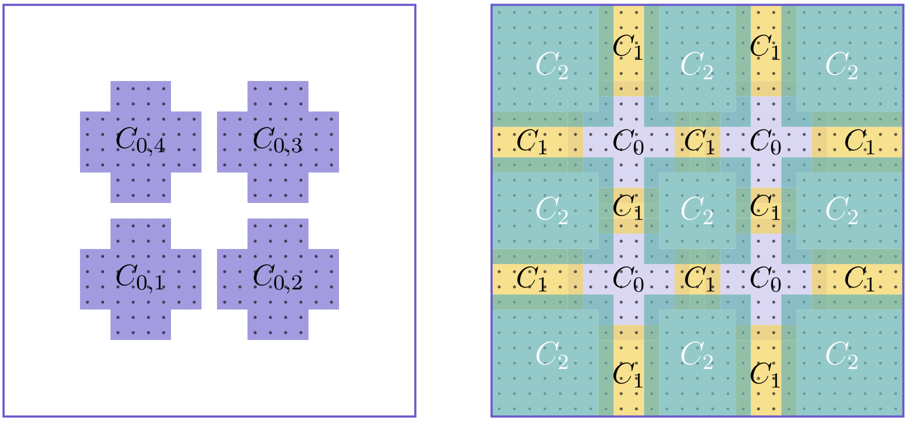

Figure 7. On the left side, we show C_0, which is defined as $C^1 \backslash \mathring{C}_1$. On the right side, we show the coarse-graining in terms of the combined C_0, C_1 and C_2. We omit the corresponding C_x^{∂} and $\mathring{C}_x$, for $x = 0, 1, 2$, for simplicity

4. $c \leq \operatorname{dist}(C^a \backslash C_a, \mathring{C}_a) = \operatorname{dist}(C^a \backslash C_a, C^a \backslash C^{a-1})\ \forall a \in [\![1, D]\!]$,
5. $2k \leq \operatorname{dist}(C_{a,i}, C_{a,j}) \quad \forall i, j \in \mathcal{I}_a, \forall a \in [\![0, D]\!]$.

For a visualization, find an example of the case $D = 2$ on the right side of Fig. 7.

Proof. 1. We proceed by induction on a. For $a = D$, each of the sets $C^{\partial}_{D,i}$ are hypercubes with side length ℓ; hence, $C_{D,i}, \mathring{C}_{D,i}$ are hypercubes of side length at most $\ell - k$ and $\ell - (k + c)$, respectively. Disjointness is clear. For $a < D$, it holds that

$$
\begin{aligned}
C^{\partial}_{a} &:= \left\{ v \in C^{a} \ : \ \exists j \in \mathbb{Z}, b \in [\![1, D]\!] : \ v + j e_b \in \overline{C^{a}} \right\} \\
&= \bigcup_{b=1}^{D} \left(C^{a} \cap \left\{ \mathring{C}_{a+1} + \mathbb{Z} e_b \right\} \right) \\
&= \bigcup_{b=1}^{D} \left(C^{a} \cap \left\{ \mathring{C}_{a+1} + \mathbb{Z}_{2(D-a)(k+c)} e_b \right\} \right).
\end{aligned}
$$

More precisely, for each $\mathring{C}_{a+1,i}$ and each canonical direction b, we construct a set $C^{\partial}_{a,j(i,b)}$ by translating $\mathring{C}_{a+1,i}$ by $2(D-a)(k+c)$ along e_b. We then define $C^{\partial}_{a,j(i,b)}$ as this translated set minus its intersection with $\mathring{C}_{a+1,i}$. We eliminate duplicates from the collection $\{C^{\partial}_{a,j(i,b)}\}_{(i,b) \in \mathcal{I}_{a+1} \times [\![1,D]\!]}$ to obtain the index set $\mathcal{I}_a$, which we use to obtain C^{∂}_{a}. The sets $C^{\partial}_{a,i}$ are mutually disjoint by construction. Each $C^{\partial}_{a,i}$ has $D - a$ sides of length $2(D - a)(k + c)$, while the remaining sides retain the length $\ell - 2(D - a)(k+c)$ inherited from the sets composing $\mathring{C}_{a+1}$. The disjointness of $C_{a,i}$ and $\mathring{C}_{a,i}$ as well as the bounds on their sizes immediately follow.

2. The disjointedness statement in 2. follows equally from the proof in 1. The statement $C^0 = \mathring{C}_0$ follows by their definitions.
3. The proof proceeds by finite induction with at most D steps, starting from $a = D$. We consider the following cases: Base case: If $x \in C_D$, we are done. Inductive step: If $x \notin C_D$, then $x \in C^{D-1} \supseteq \Lambda_L \backslash C_D$. For each subsequent step a, we have three possibilities:
   (a) If $a = 0$, then $x \notin \bigcup_{b=1}^{D} C_b$. Consequently, $x \in C^0 = C^{\partial}_0 = C_0 = \mathring{C}_0$, and we are done.
   (b) If $x \in C_a$, we are done.
   (c) If neither (a) nor (b) holds, then $x \notin \bigcup_{b=a}^{D} C_b$. Hence, $x \in \Lambda_L \backslash \bigcup_{b=a}^{D} C_b \subset C^{a-1}$, and the induction continues to the next step.

   The induction terminates after at most D steps because there are only $D+1$ sets C_a. And since each is a disjoint union of $C_{a,i}$ every $x \in \Lambda_L$ is contained in at most $D + 1$ of the $C_{a,i}$.
4. Recall that $k, c \geq r$, the range of the interactions. The claim follows via

$$
\begin{aligned}
\operatorname{dist}(C^{a} \backslash C_a, \mathring{C}_a) &= \operatorname{dist}(C^{a} \backslash C_{a,i}, \mathring{C}_{a,i}) \\
&= \operatorname{dist}(\overline{C_{a,i}} \cap C^{a}, \mathring{C}_{a,i} \cap C^{a}) \\
&= \operatorname{dist}(\{x \in C^{a} : \operatorname{dist}(x, \overline{C^{\partial}_{a,i}}) \leq k\}, \\
&\qquad \{x \in C^{a} : \operatorname{dist}(x, \overline{C^{\partial}_{a,i}}) > k + c\}) = c
\end{aligned}
$$

and it is easy to check that $C^a \backslash C^{a-1} = C^a \backslash (C^a \backslash \mathring{C}_a) = \mathring{C}_a$ for $a \in [\![1, D]\!]$.

5. This follows directly from the disjointness of $C_{a,i}$ and $C_{a,j}$, see 1. and their definition.

□

B.3. A Weak Approximate Tensorization for Davies Channels

By combining the general weak entropy factorization from Appendix B.1 with the coarse-graining in Appendix B.2, this section contains a weak approximate tensorization (wAT) for the Davies channels with corrections that are the MCMIs of the splitting. Under the assumption that the Gibbs state exhibits exponential decay of its matrix-valued quantum conditional mutual information, this correction term decays exponentially with the overlap length l (see Lemma B.2).

Theorem B.3 (Weak approximate tensorization). *Given the decomposition of the lattices Λ with constants $(k \geq r, c, l)$ described in the previous section (c.f. Appendix B.2) and then in the context of Appendix A.3, the family $\{E_{C_{a,i}} : a \in [\![0, D]\!], i \in \mathcal{I}_a\}$ of Davies conditional expectations with respect to σ satisfies the following inequality: for all $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$,*

$$D(\rho\|\sigma) \leq \sum_{a=0}^{D} \sum_{i_a \in \mathcal{I}_a} D(\rho\|E_{C_{a,i_a}}(\rho)) + \sum_{a=1}^{D} \zeta_a(\sigma)\,, \tag{53}$$

with

$$\zeta_a(\sigma) := ||\mathbf{H}_\sigma(X_a : Z_a|W_a)||_\infty\,, \tag{54}$$

and $W_a \sqcup X_a \sqcup Y_a \sqcup Z_a =: \overline{C^a} \sqcup \mathring{C}_a \sqcup (C_a \backslash \mathring{C}_a) \sqcup (C^a \backslash C_a)$ with $d(X_a, Z_a) = c \geq r$, for $a \in [\![1, D]\!]$. Assuming further that the Gibbs state satisfies exponential decay of MCMI with constants K and ξ, then we can estimate (53) with

$$D(\rho\|\sigma) \leq \sum_{a=0}^{D} \sum_{i_a \in \mathcal{I}_a} D(\rho\|E_{C_{a,i_a}}(\rho)) + DK|\Lambda_L|e^{-c/\xi}\,. \tag{55}$$

Proof. We proceed by induction on a, starting at $a = D$ and decreasing to zero. Using the notation and result from Lemma B.1, and conditioning on $W_D = \emptyset = \overline{C^D} = \overline{\Lambda}$, we obtain:

$$\begin{aligned}
D(\rho\|\sigma) &= D_{C^D}(\rho\|\sigma) \\
&= D_{X_D Y_D Z_D}(\rho\|\sigma) \leq D_{X_D Y_D}(\rho\|\sigma) + D_{Y_D Z_D}(\rho\|\sigma) \\
&\quad + ||\mathbf{H}_\sigma(X_D : Z_D|\emptyset)||_\infty \\
&= D_{C_D}(\rho\|\sigma) + D_{C^{D-1}}(\rho\|\sigma) + \zeta_D(\sigma) \\
&\overset{(1)}{\leq} D(\rho\|E_{C_D}(\sigma)) + D_{C^{D-1}}(\rho\|\sigma) + \zeta_D(\sigma) \\
&\overset{(2)}{\leq} \sum_{i \in \mathcal{I}_D} D(\rho\|E_{C_{D,i}}(\sigma)) \\
&\quad + D_{C^{D-1}}(\rho\|\sigma) + \zeta_D(\sigma).
\end{aligned}$$

In step (1), we apply (50). Then, we use the fact that $C_D = \bigsqcup_{i\in\mathcal{I}_D} C_{D,i}$ with $\operatorname{dist}(C_{i,D}, C_{j,D}) > k \geq r$, as per Lemma B.2. This allows us to apply 3. of the properties of the Davies channels from Appendix A.3, i.e split $E_D = \prod_{i\in\mathcal{I}_D} E_{C_{D,i}}$ into commuting constituents. Consequently, in step (2), we split the relative entropy according to (25). The induction proceeds with $D_{C^{D-1}}(\rho\|\sigma)$.

The induction step follows a similar line of reasoning: For $a \neq 0$, we again employ the weak entropy factorization from Lemma B.1:

$$
\begin{aligned}
D_{C^a}(\rho\|\sigma) = D_{X_aY_aZ_a}(\rho\|\sigma) &\leq D_{X_aY_a}(\rho\|\sigma) + D_{Y_aZ_a}(\rho\|\sigma) + ||H_\sigma(X_a : Z_a|W_a)||_\infty \\
&= D_{C_a}(\rho\|\sigma) + D_{C^{a-1}}(\rho\|\sigma) + ||\mathbf{H}_\sigma(X_a : Z_a|W_a)||_\infty \\
&\leq \sum_{i\in\mathcal{I}_a} D(\rho\|E_{C_{a,i}}(\rho)) + D_{C^{a-1}}(\rho\|\sigma) + \zeta_a(\sigma)\,.
\end{aligned}
$$

For $a = 0$, we only need to estimate $D_{C^0}(\rho\|\sigma) = D_{C_0}(\rho\|\sigma) \leq \sum_{i\in\mathcal{I}_0} D(\rho\|E_{C_{0,i}}(\rho))$. This inequality holds because we first can employ (50) and then use that $C_0 = \bigsqcup_{i\in\mathcal{I}_0} C_{0,i}$, where the $C_{0,i}$ have mutual distances greater than r. Thus, 3. for the Davies channels from Appendix A.3 applies, allowing us to decompose E_{C_0} into a mutually commuting composition of the $E_{C_{0,i}}$ and then apply (25). The final inequality holds because again $C_a = \bigsqcup_{i\in\mathcal{I}_a} C_{a,i}$ with $\operatorname{dist}(C_{a,i}, C_{a,j}) > r$ for $i \neq j$ by Lemma B.2, allowing us to apply property 3. for Davies channels from Appendix A.3 and, subsequently, (25). We thereby complete the induction. In Fig. 8, the decomposition of the lattice for the weak approximate tensorization is demonstrated for the case $D = 2$ and $a = D$ and $a = D - 1$, respectively.

In the case of exponential decay of matrix-valued quantum conditional mutual information, we have:

$$
\zeta_a(\sigma) \leq K|\Lambda|e^{-c/\xi}\,,
$$

independent of a, which immediately gives (55). □

B.4. A Weak Transport Cost Inequality for Davies Channels

Tailored to our weak approximate tensorization from the previous section Appendix B.3 but valid also for arbitrary coarse-graining of the lattice with Davies channels, we derive in this section a weak transport cost (wTC) inequality which forms a crucial ingredient in the proof of the Wasserstein mixing result. Let us begin with an auxiliary lemma.

Lemma B.4. *Let $\mathcal{L}$ a generator of a GNS-symmetric quantum Markov semigroup with invariant state $\sigma > 0$ that satisfies (wMLSI) with constants c_1 and c_2 and let $\rho \in \mathcal{S}(\mathcal{H})$. Assume that $D(\rho\|\sigma) \geq c_2$. Then for $t \in (0, t_{c_2}(\rho))$ where $t_{c_2}(\rho) := \sup\{t \geq 0 : D(e^{t\mathcal{L}}(\rho)\|\sigma) \geq c_2\}$*

$$
\int_0^t \sqrt{\mathrm{EP}_{\mathcal{L}}(\rho_s)}ds \leq 2\sqrt{c_1(D(\rho\|\sigma) - c_2)} \leq 2\sqrt{c_1 D(\rho\|\sigma)} \quad \textit{with} \quad \rho_s := e^{s\mathcal{L}}(\rho)\,. \tag{56}
$$

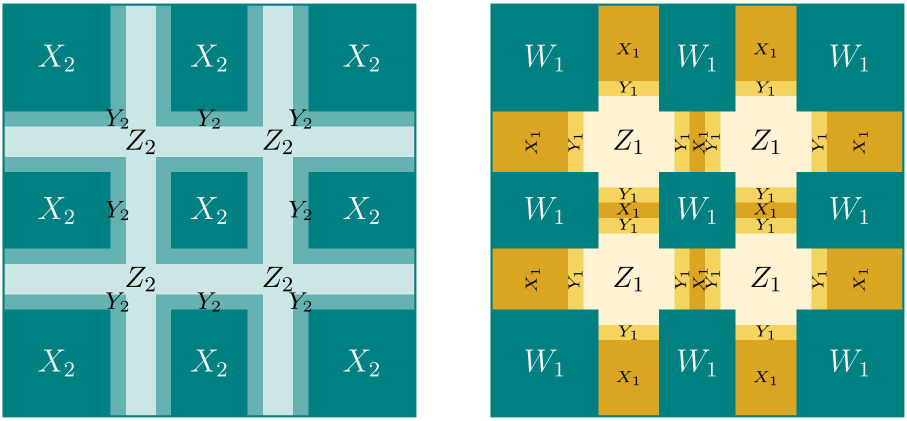

FIGURE 8. The decomposition of the lattice used in the weak approximate tensorization for the case $D = 2$ in the steps $a = 2$ on the left and $a = 1$ on the right

Proof. Let us define $F : (0, t_{c_2(\rho)}) \to \mathbb{R}$ as

$$F(t) := \int_0^t \sqrt{\mathrm{EP}_{\mathcal{L}}(\rho_s)}\mathrm{d}s + 2\sqrt{c_1(D(\rho_t\|\sigma) - c_2)}\,. \tag{57}$$

This function is non-increasing in t, as

$$\frac{\mathrm{d}}{\mathrm{d}t}F(t) = \sqrt{\mathrm{EP}_{\mathcal{L}}(\rho_t)} - \frac{\sqrt{c_1}\,\mathrm{EP}_{\mathcal{L}}(\rho_t)}{\sqrt{D(\rho_t\|\sigma) - c_2}} \leq 0 \tag{58}$$

due to wMLSI. The claim then immediately follows by positivity of the entropy production. □

With this lemma in place, we can now move our attention to the proof of the wTC from the wAT.

Theorem B.5 (Weak transport cost inequality). *In the context of Appendix A.3, assume that there exists $A = \{A_i\}_{i=1}^{n_A}$ with $A_i \subseteq \Lambda$ an overlapping coarse-graining of Λ, i.e. $\bigcup_{i=1}^{n_A} A_i = \Lambda$ with corresponding Davies projectors E_{A_i} such that one has*

$$D(\rho\|\sigma) \leq \sum_{i=1}^{n_A} D(\rho\|E_{A_i}(\rho)) + c_2 \tag{59}$$

for all $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$, then the following holds

$$||\rho - \sigma||_{W_1} \leq \max_i 2\sqrt{2}|A_i\partial|\sqrt{n_A D(\rho\|\sigma)} + |\Lambda|\sqrt{2c_2} \tag{60}$$

for all $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$.

Proof. In the following let us denote $\mathcal{L}^H_{A_i} := E_{A_i} - \mathrm{id}$. By non-negativity of the relative entropy, it holds that

$$\begin{aligned} D(\rho\|E_{A_i}(\rho)) &\leq D(\rho\|E_{A_i}(\rho)) + D(E_{A_i}(\rho)\|\rho) \\ &= \mathrm{tr}[(\mathrm{id} - E_{A_i})(\rho)(\log(\rho) - \log E_{A_i}(\rho))] \\ &= \mathrm{tr}[(\mathrm{id} - E_{A_i})(\rho)(\log(\rho) - \log \sigma)] \\ &= \mathrm{EP}_{\mathcal{L}^H_{A_i}}(\rho)\,, \end{aligned} \tag{61}$$

where in the last equality we used that $\log E_{A_i}(\rho) - \log\sigma$ is a fixed point of E^*_A which immediately gives $\mathrm{tr}[(\mathrm{id} - E_{A_i})(\rho)(\log E_{A_i}(\rho) - \log \sigma)] = 0$. Using (61), the assumption can be rewritten as

$$D(\rho\|\sigma) \leq \mathrm{EP}_{\mathcal{L}^H_\Lambda}(\rho) + c_2 \tag{62}$$

where we defined the primitive GNS-symmetric generator $\mathcal{L}^H_\Lambda := \sum_{i=1}^{n_A} \mathcal{L}^H_{A_i}$. As E_{A_i} only acts non-trivially on $A_i\partial$, we find that $\mathrm{tr}_{A_i\partial} \circ \mathcal{L}^H_{A_i} = 0$ which is a direct consequence of (37). From this fact, one immediately obtains

$$\left|\left|\mathcal{L}^H_{A_i}(\rho)\right|\right|_{W_1} \leq |A_i\partial| \sqrt{2\,\mathrm{EP}_{\mathcal{L}^H_{A_i}}(\rho)}\,, \tag{63}$$

as by (23) one has

$$\left|\left|\mathcal{L}^H_{A_i}(\rho)\right|\right|_{W_1} \leq |A_i\partial| \left|\left|\mathcal{L}^H_{A_i}(\rho)\right|\right|_1 .$$

Now using Pinskers' inequality we get $\left|\left|\mathcal{L}^H_{A_i}(\rho)\right|\right|_1 \leq \sqrt{2D(\rho\|E_{A_i}(\rho))}$ which combined with (61) gives (63). We can now shift our attention to the main result. First assume that $D(\rho\|\sigma) \leq c_2$, then the inequality holds trivially by $||\cdot||_{W_1} \leq |\Lambda|\, ||\cdot||_1$ (63) and Pinsker's inequality (19). If this is not the case we set $t_{c_2} := \sup\{t \geq 0 : D(e^{t\mathcal{L}^H_\Lambda}(\rho)\|\sigma) \geq c_2\}$ and get

$$||\rho - \sigma||_{W_1} \leq \left|\left|\rho - \rho_{t_{c_2}}\right|\right|_{W_1} + \left|\left|\rho_{t_{c_2}} - \sigma\right|\right|_{W_1} \leq \left|\left|\rho - \rho_{t_{c_2}}\right|\right|_{W_1} + |\Lambda|\sqrt{2c_2}$$

where we defined $\rho_t := e^{t\mathcal{L}^H_\Lambda}(\rho)$ for $t \geq 0$. The second inequality follows again by the above reasoning. For the first term, we write

$$\left|\left|\rho - \rho_{t_{c_2}}\right|\right|_{W_1} \leq \int_0^{t_{c_2}} \left|\left|\mathcal{L}^H_\Lambda(\rho_s)\right|\right|_{W_1} \mathrm{d}s \leq \int_0^{t_{c_2}} \sum_{i=1}^{n_A} \left|\left|\mathcal{L}^H_{A_i}(\rho_s)\right|\right|_{W_1} \mathrm{d}s\,.$$

By (63), we get

$$\int_0^{t_{c_2}} \sum_{i=1}^{n_A} \left|\left|\mathcal{L}^H_{A_i}(\rho_s)\right|\right|_{W_1} \mathrm{d}s \leq \max_i \sqrt{2}|A_i\partial| \int_0^{t_{c_2}} \sum_{i=1}^{n_A} \sqrt{\mathrm{EP}_{\mathcal{L}^H_{A_i}}(\rho_s)}\mathrm{d}s$$

and lastly through concavity of $x \mapsto \sqrt{x}$,

$$\left|\left|\rho - \rho_{t_{c_2}}\right|\right|_{W_1} \leq \max_i \sqrt{2}|A_i\partial|\sqrt{n_A} \int_0^{t_{c_2}} \sqrt{\mathrm{EP}_{\mathcal{L}^H_\Lambda}(\rho_s)}\mathrm{d}s\,.$$

Using (62), the claim follows by Lemma B.4. $\square$

B.5. A MLSI Alike Inequality for Local Davies Semigroups at Every Temperature

This section is dedicated to collecting some results about Davies conditional expectations and entropy productions that will all be used in an inequality that almost resembles an MLSI for the local Davies Lindbladians. More precisely, the following inequality

$$D(\rho\|E_A(\rho)) \leq \mathcal{O}(e^{\mathcal{O}(|A\partial|)})\,\mathrm{EP}_{\mathcal{L}_{A\partial}}(\rho)\,, \tag{64}$$

which will be shown and used in the next section, i.e. Appendix B.6. As our proof will rely on going through the infinite temperature Davies (i.e. $\lim_{\beta\to 0}\mathcal{L}_A^\beta = \mathcal{L}_A^0$), we will keep the temperature explicit in the following. The strategy will be to relate the LHS and the entropy production on the RHS of (64) to their infinite temperature counterparts through Lemma B.6 and then to proof (64) only for the infinite temperature Davies, where the semigroup is almost the depolarizing one.

Lemma B.6. *In the context of Appendix A.3, let $A \subseteq \Lambda$ and define $E_A^\beta = \lim_{t\to\infty} e^{t\mathcal{L}_A^{D,\beta}}$ and $E_A^0 = \lim_{t\to\infty} e^{t\mathcal{L}_A^{D,0}}$. Then the following inequalities hold:*

$$D(\rho\|E_A^\beta(\rho)) \leq e^{2gJ\beta|A\partial|}D(\rho\|E_A^0(\rho))\,. \tag{65}$$

and

$$\mathrm{EP}_{\mathcal{L}_A^{D,0}}(\rho) \leq e^{2gJ(\beta|A\partial|+1)}\,\mathrm{EP}_{\mathcal{L}_A^{D,\beta}}(\rho) \tag{66}$$

Proof. Let $\sigma = E_A^\beta(d_\Lambda^{-1}I_\Lambda)$ and $\sigma' = E_A^0(d_\Lambda^{-1}I_\Lambda) = d_\Lambda^{-1}I_\Lambda$. Then $\mathcal{L}_A^{D,\beta}$ is GNS-symmetric with respect to σ, and $\mathcal{L}_A^{D,0}$ is GNS-symmetric with respect to σ'. Applying [58, Proposition 4.2], we obtain:

$$D(\rho\|E_A^\beta(\rho)) \leq \|E_A^\beta(I_\Lambda)\|_\infty D(\rho\|E_A^0(\rho))\,.$$

By positivity preservation of the conditional expectation and definition, we have

$$E_A^\beta(I_\Lambda) \leq \|e^{\beta H_{A\partial}}\|_\infty E_A^\beta(e^{-\beta H_{A\partial}}) = \|e^{\beta H_{A\partial}}\|_\infty e^{-\beta H_{A\partial}}, \tag{67}$$

and hence, $\|E_A^\beta(I_\Lambda)\|_\infty \leq \left|\left|e^{\beta H_{A\partial}}\right|\right|_\infty \left|\left|e^{-\beta H_{A\partial}}\right|\right|_\infty \leq e^{2\beta gJ|A\partial|}$. For the entropy productions, [58, Proposition 4.3] yields:

$$\mathrm{EP}_{\mathcal{L}_A^{D,0}}(\rho) \leq \max_\omega e^{|\omega|/2}\|(E_A^\beta(I_\Lambda))^{-1}\|_\infty\,\mathrm{EP}_{\mathcal{L}_A^{D,\beta}}(\rho) \tag{68}$$

The Bohr frequencies are localized and can be bounded above by $|\omega| \leq \max_{k\in A} 2\,||H_{k\partial}||_\infty \leq 2gJ$ as per (35). Furthermore, $\|(E_A^\beta(I_\Lambda))^{-1}\|_\infty \leq \left|\left|e^{\beta H_{A\partial}}\right|\right|_\infty \leq \left|\left|e^{-\beta H_{A\partial}}\right|\right|_\infty \leq e^{2\beta gJ|A\partial|}$, completing the proof. $\square$

To conclude, let us present (64) for the infinite temperature Davies.

Lemma B.7. *In the context of Appendix A.3, let $A \subseteq \Lambda$ and $E_A^0 = \lim_{t\to\infty} e^{t\mathcal{L}_A^{D,0}}$. Then:*

$$\chi_{\min}^0 D(\rho\|E_A^0(\rho)) \leq \mathrm{EP}_{\mathcal{L}_{A\partial}^{D,0}}(\rho)\,. \tag{69}$$

Proof. By the GNS symmetry of $\mathcal{L}^0_{A\partial}$, we can employ [32, Theorem 5.10] to express $\mathrm{EP}_{\mathcal{L}^{D,0}_{A\partial}}(\rho)$ as a sum of non-negative inner products with positive coefficients, including $\chi^0_{\omega,k}$. Using the bound $\chi^0_{\omega,k} \geq \chi^0_{\min}$, we obtain:

$$\chi^0_{\min}\,\mathrm{EP}_{\tilde{\mathcal{L}}^{D,0}_{A\partial}} \leq \mathrm{EP}_{\mathcal{L}^{D,0}_{A\partial}}\,,$$

where

$$\begin{aligned}\tilde{\mathcal{L}}^{D,0}_{A\partial} &= \sum_{k\in A\partial}\sum_{\omega,\alpha}\left(S^{\omega}_{\alpha,k}\rho S^{\omega,\dagger}_{\alpha,k} - \frac{1}{2}\left\{\rho, S^{\omega,\dagger}_{\alpha,k}S^{\omega}_{\alpha,k}\right\}\right)\\ &= \sum_{t\geq 0}\sum_{k\in A\partial}\left(\sum_{\alpha}\Delta_{e^{itH}}\left(S_{\alpha,k}\Delta_{e^{-itH}}(\rho)S_{\alpha,k}\right) - \rho\right).\end{aligned}$$

The last simplification follows from [60, Theorem 31]. Define $\rho_t := \Delta_{e^{-itH}}(\rho)$. Using the gauge freedom of entropy production with respect to the choice of fixed point, and noting that I_Λ/d_Λ is a fixed point, we have:

$$\begin{aligned}\mathrm{EP}_{\tilde{\mathcal{L}}^{D,0}_{A\partial}}(\rho) &= -\mathrm{tr}[\tilde{\mathcal{L}}^{D,0}_{A\partial}(\rho)(\log(\rho) - \log I/d)]\\ &= \mathrm{tr}[\tilde{\mathcal{L}}^{D,0}_{A\partial}(\rho)\log(\rho)]\\ &= \sum_{t\geq 0}\sum_{k\in A\partial,}\mathrm{tr}[(\rho_t - \sum_{\alpha}S_{\alpha,k}\rho_t S_{\alpha,k})\log(\rho_t)]\\ &= \sum_{t\geq 0}\mathrm{EP}_{\mathcal{L}^{\mathrm{depol}}_{A\partial}}(\rho_t)\\ &\geq \min_t \mathrm{EP}_{\mathcal{L}^{\mathrm{depol}}_{A\partial}}(\rho_t)\end{aligned}$$

where $\mathcal{L}^{\mathrm{depol}}_{A\partial}(\,\cdot\,) = \sum_{k\in A\partial}(\sum_{\alpha}S_{\alpha,k}(\,\cdot\,)S_{\alpha,k} - \mathrm{id}) = \sum_{k\in A\partial}(\mathrm{tr}_k[\,\cdot\,] - \mathrm{id})$. Since the depolarizing channel has a cMLSI of 1 [29], and using the chain rule of relative entropy (24), we have:

$$\begin{aligned}\mathrm{EP}_{\mathcal{L}^{\mathrm{depol}}_{A\partial}}(\rho_t) &\geq D(\rho_t\|\mathbb{E}_{A\partial}(\rho_t)) = D(\rho_t\|d_\Lambda^{-1}I_\Lambda) - D(\mathbb{E}_{A\partial}(\rho_t)\|d_\Lambda^{-1}I_\Lambda)\\ &= D(\rho_t\|d_\Lambda^{-1}I_\Lambda) - D(\mathbb{E}_{A\partial}(E^0_A(\rho_t))\|\mathbb{E}_{A\partial}(d_\Lambda^{-1}I_\Lambda))\\ &\overset{\mathrm{DPI}}{\geq} D(\rho_t\|d_\Lambda^{-1}I_\Lambda) - D(E^0_A(\rho_t)\|d_\Lambda^{-1}I_\Lambda)\\ &= D(\rho_t\|E^0_A(\rho_t))\end{aligned}$$

We used that $\mathbb{E}_{A\partial}\circ E_A = \mathbb{E}_{A\partial}$ (which follows from $\mathbb{E}_{A\partial}\circ\mathcal{A}_{A,\sigma} = \mathbb{E}_{A\partial}$ and (36)), the data processing inequality (DPI) and the chain rule. Finally, observe that $[E^0_A, \Delta_{e^{-itH}}] = 0$ by the GNS symmetry of $\mathcal{L}^\beta_A$ which is preserved in the limit, therefore:

$$D(\rho_t\|E^0_A(\rho_t)) = D(\Delta_{e^{-itH}}(\rho)\|\Delta_{e^{-itH}}(E^0_A(\rho))) = D(\rho\|E^0_A(\rho))\,.$$

□

B.6. Weak MLSI for the Global Davies Semigroup

In this section, we are going to show two weak MLSIs for the global Davies semigroups relying in one case solely on the MCMI decay (c.f. Theorem B.8), while in the other case (c.f. Theorem B.10), we replace the result from Appendix B.5 with an assumption on the local gap of the generator to improve the scaling with system size. Hence, the second result relies on the existence of MCMI and a uniformly polynomial local gap. More precisely, we require the following *uniform local polynomial gap*: There exists $\mu \in \mathbb{N}_0$ and $C > 0$ such that for all $A \subseteq \Lambda$

$$\lambda(\mathcal{L}_A) \geq C|A|^{-\mu} = \Omega(|A|^{-\mu})\,. \tag{70}$$

The first result will be then used in Sect. C to derive the quasi-rapid Wasserstein mixing under every temperature while the second one improves the scaling quasi-rapid Wasserstein mixing and further gives rapid mixing in trace distance under the assumption of a polynomial local gap. Let us begin with the first result only requiring MCMI decay and no additional assumptions on the local or global gap.

Theorem B.8 (weak MLSI at every β). *In the setting of Appendix A.3 assume that the Gibbs state at inverse temperature β satisfies MCMI decay with constants K, ξ, then the semigroup $\{e^{t\mathcal{L}_\Lambda^D}\}_{t\geq 0}$ at that temperature satisfies the following weak MLSI for $\epsilon \geq KD2^D(2\,L+1)^D\exp(\frac{2r}{\xi}-\frac{L}{D\xi}) = \mathcal{O}(L^D e^{-\mathcal{O}(L)})$.*

$$D(\rho_t\|\sigma) \leq e^{-\alpha(\epsilon)t}D(\rho\|\sigma) + \epsilon \tag{71}$$

with $\frac{1}{\alpha(\epsilon)} = (D+1)(\chi^0_{\min})^{-1}e^{2gJ}e^{4\beta gJ(2D(2r+\xi\log\frac{KD2^DN}{\epsilon})+1)^D} = \mathcal{O}(\exp\{\mathcal{O}((\log\frac{N}{\epsilon})^D)\})$, *where $\rho_t = e^{t\mathcal{L}_\Lambda^D}(\rho)$ for arbitrary $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$.*

Such a lower bound on the admissible ϵ is generic and to be expected. Likewise we can also interpret this as giving a minimal lattice size N for which this MLSI eventually holds for a given fixed ϵ.

Proof. Using Theorem B.3 w.r.t. a coarse-graining with parameters $(k = 2r - \frac{r}{D}, c, l)$ and both Lemmas B.6 and B.7 yields

$$\begin{aligned}
D(\rho\|\sigma) &\leq \sum_{a,i} D(\rho\|E^D_{C_{a,i}}(\rho)) + D2^DKNe^{-\frac{c}{\xi}} \\
&\leq \sum_{a,i} e^{2\beta gJ|C_{a,i}\partial|}e^{2\beta gJ|C_{a,i}\partial\partial|}e^{2gJ}(\chi^0_{\min})^{-1}\,\mathrm{EP}_{\mathcal{L}^D_{C_{a,i}\partial}}(\rho) + D2^DKNe^{-\frac{c}{\xi}} \\
&\leq (D+1)(\chi^0_{\min})^{-1}e^{4\beta gJ(\ell+2r)^D+2gJ}\,\mathrm{EP}_{\mathcal{L}^D_\Lambda}(\rho) + D2^DKNe^{-\frac{c}{\xi}},
\end{aligned}$$

where we used that $|C_{a,i}\partial| \leq |C_{a,i}\partial\partial| \leq (\ell+2r)^D$ by construction of this coarse-graining (c.f. Lemma B.2). Picking $c = \max\{\xi\log\frac{KD2^DN}{\epsilon}, r\} = \mathcal{O}(\log\frac{N}{\epsilon})$ and $\ell = 2D(k+c)+1 = 2D(2r+c)+1-2r = \mathcal{O}(\log\frac{N}{\epsilon})$ gives a valid coarse-graining iff $c, \ell = 2D(k+c)+1 \leq 2L+1$, since $r \leq k \leq 2r \leq 2L+1$.

This is equivalent to $c \leq \frac{L}{D} - k$, which is implies by $c \leq \frac{L}{D} - 2r$, and thus, $\epsilon \geq KD2^D N \exp(\frac{2r}{\xi} - \frac{L}{D\xi}) \geq KD2^D(2\,L+1)^D \exp(\frac{2r}{\xi} - \frac{L}{D\xi})$. We get

$$D(\rho\|\sigma) \leq \frac{(D+1)e^{2gJ}}{\chi^0_{\min}} e^{4\beta gJ(2D(2r+\xi\log\frac{KD2^D N}{\epsilon})+1)^D}\, \mathrm{EP}_{\mathcal{L}_\Lambda}(\rho) + \epsilon \tag{72}$$

$$= \mathcal{O}\left(\exp(\mathcal{O}\left(\left(\log\frac{N}{\epsilon}\right)^D\right))\right) \mathrm{EP}_{\mathcal{L}_\Lambda}(\rho) + \epsilon. \tag{73}$$

Applying Grönwalls lemma now yields the desired statement. □

For the second result, let us first convert the uniform polynomial local gap into a uniform polynomial cMLSI of the local generators $\mathcal{L}_A$:

Lemma B.9. *In the context of Appendix A.3 a uniform polynomial lower bound on the local gap with constants C and μ implies a uniform lower bound on the cMLSI constant of the form*

$$\alpha_c(\mathcal{L}_A) \geq \frac{C}{2\log 10 + 2(2\beta gJ + 3\log d)|A\partial|}|A|^{-\mu} \geq \Omega(|A|^{-\mu-1})\,. \tag{74}$$

Proof. The bound is a consequence of (30) and a suitable bound on the Pimsner–Popa index in the setting of local Davies generators. Let us denote with $E_A = \lim_{t\to\infty} e^{t\mathcal{L}_A}$. In [51], the authors showed that

$$C_{cb}(E_A) \leq \left|\left|\tau^{-1}\right|\right|_\infty \sum_{i=1}^n d^2_{\mathcal{K}_i}\,, \tag{75}$$

where

$$E_A(\rho) = \bigoplus_{i=1}^n \mathrm{tr}_{\mathcal{K}_i}[P_i\rho P_i] \otimes \tau_i \quad \text{and} \quad \tau = \bigoplus_{i=1}^n I_{\mathcal{H}_i} \otimes \tau_i\,. \tag{76}$$

is the decomposition of the Davies channel. As E_A only acts non-trivially on $A\partial$, we can conclude that $\sum_{i=1}^n d_{\mathcal{K}_i} \leq d_{A\partial} = d^{|A\partial|}$. By superadditivity of $x \mapsto x^2$, we hence get $\sum_{i=1}^n d^2_{\mathcal{K}_i} \leq \left(\sum_{i=1}^n d_{\mathcal{K}_i}\right)^2 \leq d^{2|A\partial|}$. To estimate $\left|\left|\tau^{-1}\right|\right|$, note that

$$E_A(d^{-1}_{A\partial} I_\Lambda) = \bigoplus_{i=1}^n \frac{d_{\mathcal{K}_i}}{d_{A\partial}} I_{\mathcal{H}_i} \otimes \tau_i \leq \bigoplus_{i=1}^n I_{\mathcal{H}_i} \otimes \tau_i = \tau\,,$$

hence $\left|\left|\tau^{-1}\right|\right|_\infty \leq d^{|A\partial|} \left|\left|(E_A(I_\Lambda))^{-1}\right|\right|_\infty \leq d^{|A\partial|} e^{2\beta gJ|A\partial|}$. The last inequality follows by the same argument as in the proof of Lemma B.6. □

Theorem B.10 (weak MLSI under polynomial gap). *In the setting of Appendix A.3 assume that the local Davies generators satisfy the polynomial local gap assumption for some $C > 0, \mu \in \mathbb{N}_0$ and further that the Gibbs state satisfies MCMI decay with constants K, ξ both for a fixed temperature β, then the semigroup $\{e^{t\mathcal{L}^D_{\Lambda_L}}\}_{t\geq 0}$ at that temperature satisfies the following weak MLSI for $\epsilon \geq KD2^D(2\,L+1)^D \exp(\frac{r}{\xi} - \frac{L}{D\xi}) = \mathcal{O}(L^D e^{-\mathcal{O}(L)})$.*

$$D(e^{t\mathcal{L}^D_{\Lambda_L}}(\rho)\|\sigma) \leq e^{-\alpha(\epsilon)t} D(\rho\|\sigma) + \epsilon \tag{77}$$

with $\frac{1}{\alpha(\epsilon)} \leq \frac{D+1}{C}(5 + 4\beta gJ + 8\log d)\left(2D\left(r + \xi\log\frac{KD^2DN}{\epsilon}\right)\right)^{D(1+\mu)} = \mathcal{O}\left(\left(\log\frac{N}{\epsilon}\right)^{D(1+\mu)}\right)$, *where* $\rho_t = e^{t\mathcal{L}^D_{\Lambda_L}}(\rho)$ *for arbitrary* $\rho \in \mathcal{S}(\mathcal{H}_\Lambda)$*.*

Remark 5. Note that instead of a uniform polynomial local gap, one can also require very high temperature, i.e. $\beta \sim \frac{1}{N}$ leading to a constant correction in Lemma B.6, giving a result analogous to the one above. This means at such very high temperatures, one gets the strengthened Theorem C.2 only from the uniform decay of MCMI.

Proof of Theorem B.10. We fix a valid coarse-graining with constants (k, c, l) to be chosen later. Then by Theorem B.3 and the MCMI decay assumed in the statement of the theorem, we have

$$\begin{aligned}
D(\rho\|\sigma) &\leq \sum_{a,i} D(\rho\|E_{C_{a,i}}(\rho)) + KD2^D Ne^{-\frac{c}{\xi}} \\
&\leq \sum_{a,i} \frac{1}{\alpha(\mathcal{L}_{C_{a,i}})} \operatorname{EP}_{\mathcal{L}_{C_{a,i}}}(\rho) + KD2^D Ne^{-\frac{c}{\xi}} \\
&\leq \frac{D+1}{C}[2\log 10 + (4\beta gJ + 6\log d)\ell^D]\ell^{D\mu} \operatorname{EP}_{\mathcal{L}_\Lambda}(\rho) + KD2^D Ne^{-\frac{c}{\xi}}.
\end{aligned}$$

Now choosing $k = r, c = \xi\log\frac{KD2^DN}{\epsilon} = \mathcal{O}\left(\log\frac{N}{\epsilon}\right)$, and $\ell = 2D(r+c)+1 = \mathcal{O}\left(\log\frac{N}{\epsilon}\right)$ gives a valid coarse-graining iff $c, \ell = 2D(k+c)+1 \leq 2L+1$. This is equivalent to $c \leq \frac{L}{D} - r$, and thus, $\epsilon \geq KD2^DN\exp(-\frac{c}{\xi}) \geq KD2^D(2L+1)^D\exp(\frac{r}{\xi} - \frac{L}{D\xi})$. We get

$$D(\rho\|\sigma) \leq \alpha(\epsilon)\operatorname{EP}_{\mathcal{L}_\Lambda}(\rho) + \epsilon,$$

where

$$\begin{aligned}
\frac{1}{\alpha(\epsilon)} &= \frac{D+1}{C}[2\log 10 + (4\beta gJ + 6\log d)\ell^D]\ell^{D\mu} \\
&\leq \frac{D+1}{C}\ell^{D(1+\mu)}(5 + 4\beta gJ + 6\log d) \\
&= \frac{D+1}{C}(5 + 4\beta gJ + 6\log d)\left(2D(r + \xi\log\frac{KD2^DN}{\epsilon}) + 1\right)^{D(1+\mu)} \\
&= \mathcal{O}\left(\left(\log\frac{N}{\epsilon}\right)^{D(1+\mu)}\right),
\end{aligned}$$

where we used that $\ell \geq 1$ and $\max\{2\log 10, 6\log d\} < 8\log d$ with $d \geq 2$. Integration now gives the statement of the theorem. □

C. Main Results—Extended

C.1. W_1-Mixing from MCMI Decay

Here we state and prove the main result of this work concerning the quasi-rapid W_1-decay.

Theorem C.1 (Quasi-rapid Wasserstein mixing). *Let $\mathcal{L}_\Lambda^D$ be a Davies Lindbladian at inverse temperature $\beta > 0$ corresponding to a $(\kappa, r)-$local, J-bounded, commuting Hamiltonian H_Λ. Denote the growth constant of H_Λ on $\Lambda = [\![-L, L]\!]^D$ with g. Then if the Gibbs state of H_Λ with the same temperature (the invariant state of $\mathcal{L}_\Lambda^D$) satisfies uniform exponential decay of the MCMI with constants $K, \xi > 0$ the semigroup generated by $\mathcal{L}_\Lambda^D$ is quasi-rapidly mixing in normalized W_1-distance with mixing time*

$$
\begin{aligned}
t_{\text{mix}}^{W_1}(\varepsilon) &= \frac{(D+1)e^{2gJ}}{\chi_{\min}^0} \\
&\quad \times \exp\Bigg(4\beta g J\Bigg[2D\Bigg(2r + \xi \log \\
&\quad \left\{\frac{64KD(D+1)2^D}{\varepsilon^2}\left(2D\left(r+\xi\log\frac{8D2^DKN}{\varepsilon^2}\right)+1\right)^{2D}\right\}\Bigg)+1\Bigg]^D\Bigg) \\
&\quad \times \log\left\{\frac{64(2\beta gJ+\log d)(D+1)}{\varepsilon^2}\left(2D\left(r+\xi\log\left\{\frac{8NKD2^D}{\varepsilon^2}\right\}\right)+1\right)^{2D}\right\} \\
&= \mathcal{O}\left(\exp\left(\text{poly}\left(\log\left(\frac{1}{\varepsilon^2}\log\frac{N}{\varepsilon^2}\right)\right)\right)\right) \\
&= \text{quasi-poly}\left(\frac{1}{\varepsilon^2}\log\frac{N}{\varepsilon^2}\right) = \text{quasi-poly}(\varepsilon^{-1})_{\varepsilon\to 0}\text{quasi-log}(N)_{N\to\infty}
\end{aligned}
$$

whenever $\varepsilon \geq 8N\sqrt{(D+1)KD2^D}\exp(\frac{r}{\xi} - \frac{N^{\frac{1}{D}}-1}{4D\xi}) = \mathcal{O}\left(N\exp(-\mathcal{O}(\sqrt[D]{N}))\right)$ *is fixed.*

Remark 6. Note that the requirement of $\epsilon \geq \mathcal{O}\left(N\exp(-\mathcal{O}(\sqrt[D]{N}))\right)$ is not of relevance in implying quasi-rapid mixing, since the property of quasi-rapid mixing is only determined by the asymptotic scaling of the mixing time $t_{\text{mix}}^{W_1}(\varepsilon)$ in the system size, for any fixed ε. The asymptotics are, however, not affected by this requirement since they hold eventually for any fixed $\varepsilon > 0$ as $\lim_{N\to\infty}\mathcal{O}(N\exp(-\mathcal{O}(\sqrt[D]{N}))) = 0$. It is also a crude bound on the actual tight one scaling as $\varepsilon \geq \mathcal{O}\left((\log N)^D\exp(-\mathcal{O}(\sqrt[D]{N}))\right)$, as can be seen in the proof.

The proof will follow from the combined application of Theorems B.5 and B.8, taking care of the explicit constants along the way.

Proof of Theorem C.1. We set $N := |\Lambda|$ in the following. First, we choose a valid (k, c, ℓ) coarse-graining w.r.t which the weak AT Theorem B.3 implies the following weak TC due to Theorem B.5.

$$
\begin{aligned}
\|\rho_t - \sigma\|_{W_1} &\leq \max_{a,i} 2\sqrt{2}|C_{a,i}|\sqrt{N(D+1)D(\rho_t\|\sigma)} + N\sqrt{2D2^DKN}e^{-\frac{c}{2\xi}} \\
&\leq 2\sqrt{2}\ell^D\sqrt{D+1}\sqrt{ND(\rho_t\|\sigma)} + N\sqrt{N}\sqrt{2D2^DK}e^{-\frac{c}{2\xi}}.
\end{aligned}
$$

Now choosing $k = r, c = \xi\log\frac{2D2^DKN}{\delta^2}$ and $\ell = 2D(r+c)+1 = 2D\left(r+\xi\log\frac{2D2^DKN}{\delta^2}\right)+1 = \mathcal{O}\left(\log\frac{N}{\delta^2}\right)$, yields

$$
\|\rho_t - \sigma\|_{W_1} \leq 2\sqrt{2(D+1)}\sqrt{Nl^{2D}D(\rho_t\|\sigma)} + N\delta.
$$

Inserting the following weak MLSI from Theorem B.8, which is implied by the uniform MCMI decay:

$$D(\rho_t\|\sigma) \leq e^{-\alpha(N\ell^{-2D}\epsilon)t} D(\rho\|\sigma) + \frac{N}{\ell^{2D}}\epsilon \leq N e^{-\alpha(N\ell^{-2D}\epsilon)t}(2\beta gJ + \log d) + \frac{N}{\ell^{2D}}\epsilon,$$

where

$$\begin{aligned}
\frac{1}{\alpha(N\ell^{-2D}\epsilon)} &= \frac{D+1}{\chi^0_{\min}} e^{2gJ} e^{4\beta gJ\left[2D\left(2r+\xi\log\frac{KD2^D\ell^{2D}}{\epsilon}\right)+1\right]^D} \\
&= \frac{D+1}{\chi^0_{\min}} e^{2gJ} e^{4\beta gJ\left[2D\left(2r+\xi\log\frac{KD2^D\left(2D\left(r+\xi\log\frac{2D2^DKN}{\delta^2}\right)+1\right)^{2D}}{\epsilon}\right)+1\right]^D}.
\end{aligned}$$

Combining this with the above weak MLSI yields

$$\|\rho_t - \sigma\|_{W_1} \leq 2\sqrt{2}\sqrt{D+1}N\sqrt{e^{-\alpha(N\ell^{-2D}\epsilon)t}\ell^{2D}(2\beta gJ + \log d) + \epsilon} + N\delta\,.$$

So for times t larger than

$$\begin{aligned}
&\frac{1}{\alpha(N\ell^{-2D}\epsilon)}\log\frac{(2\beta gJ+\log d)\ell^{2D}}{\epsilon} \\
&= \frac{D+1}{\chi^0_{\min}} e^{2gJ} e^{4\beta gJ\left[2D\left(2r+\xi\log\left\{\frac{KD2^D}{\epsilon}\left(2D\left(r+\xi\log\frac{2D2^DKN}{\delta^2}\right)+1\right)^{2D}\right\}\right)+1\right]^D} \\
&\quad\times\log\left\{\frac{(2\beta gJ+\log d)}{\epsilon}\left(2D\left(r+\xi\log\frac{2D2^DKN}{\delta^2}\right)+1\right)^{2D}\right\} \\
&= \mathcal{O}\left(\exp\left(\mathrm{poly}_D\left(\log\frac{1}{\epsilon}\log\frac{N}{\delta^2}\right)\right)\right),
\end{aligned}$$

it holds that

$$\|\rho_t - \sigma\|_{W_1} \leq 2\sqrt{2(D+1)}N\sqrt{2\epsilon} + N\delta \leq N\varepsilon,$$

where we set $\epsilon := \frac{\delta^2}{16(D+1)}$ and $\delta = \frac{\varepsilon}{2}$ completes the proof with the mixing time claimed in the statement of the theorem. The bound on the minimal value of comes from the one in Theorem B.8, rewritten in terms of $N = (2L+1)^D$, yielding

$$\begin{aligned}
\delta &\geq \sqrt{2D2^DKN}\exp\left(\frac{r}{2\xi} - \frac{\sqrt[D]{N}-1}{4D\xi}\right) \\
\epsilon &\geq KD2^D\ell^{2D}\exp\left(\frac{2r}{\xi} - \frac{\sqrt[D]{N}-1}{2D\xi}\right).
\end{aligned}$$

Crudely bounding $\ell^D \leq N$ and $\varepsilon = 8\sqrt{D+1\epsilon} = 2\delta$ yields the result, i.e. it is easy to check that the in the theorem claimed bound on ε satisfies both the above inequalities. $\square$

C.2. Rapid Mixing from MCMI Decay and Polynomial Local Gap

Theorem C.2 (Rapid mixing and hyperrapid W_1 mixing under polynomial local gap). *Let $\mathcal{L}_\Lambda^D$ be a Davies Lindbladian corresponding to a (κ, r)−local, J-bounded, commuting Hamiltonia H_Λ, to inverse temperature $\beta > 0$. Denote the growth constant of H_Λ on $\Lambda = [\![-L, L]\!]^D$ with g. Then if the Gibbs state of H_Λ to inverse temperature β (the invariant state of $\mathcal{L}_\Lambda^D$) satisfies uniform exponential decay of the MCMI with constants $K, \xi > 0$ and the gap of the local Davies generators are at most polynomially decaying in local region size with degree $\mu \in \mathbb{N}_0$, the semigroup generated by $\mathcal{L}_\Lambda$ is rapidly mixing in trace distance with mixing time*

$$\begin{aligned} t^1_{mix}(\varepsilon) &= \frac{D+1}{C}\,(5 + 4\beta g J + 8\log d) \\ &\quad\times \left(2D\left(r + \xi \log \frac{2KD^2DN}{\varepsilon^2}\right)\right)^{D(1+\mu)} \\ &\quad\times \log \frac{4(2\beta gJ + \log d)N}{\varepsilon^2} \\ &= \mathcal{O}\left(\left(\log \frac{N}{\varepsilon^2}\right)^{1+D(1+\mu)}\right) \end{aligned}$$

whenever $\varepsilon \geq \sqrt{2KD2^DN}\exp(\frac{r}{2\xi} - \frac{\sqrt[D]{N}-1}{4D\xi}) = \mathcal{O}(\sqrt{N}\exp(-\mathcal{O}(\sqrt[D]{N})))$. And it is even faster than rapidly mixing in normalized W_1 distance with mixing time

$$\begin{aligned} t^{W_1}_{mix}(\varepsilon) &= \frac{D+1}{C}(5 + 4\beta gJ + 8\log d) \\ &\quad\times \left(2D\left(r + \xi \log\left(\frac{64KD2^D}{\varepsilon^2}\left(2D(r + \xi \log \frac{8D2^DKN}{\varepsilon^2}) + 1\right)^{2D}\right)\right) + 1\right)^{D(1+\mu)} \\ &\quad\times \log\left(\frac{64(2\beta gJ + \log d)}{\epsilon^2}\left(2D(r + \xi \log \frac{8D2^DKN}{\varepsilon^2}) + 1\right)^{2D}\right) \\ &= \mathcal{O}\left(\left(\log\left(\frac{1}{\varepsilon^2}\log\frac{N}{\varepsilon^2}\right)\right)^{1+D(1+\mu)}\right) \end{aligned}$$

whenever $\varepsilon \geq 8N\sqrt{(D+1)KD2^D}\exp(\frac{r}{2\xi} - \frac{N^{\frac{1}{D}}-1}{4D\xi}) = \mathcal{O}\left(N\exp(-\mathcal{O}(\sqrt[D]{N}))\right)$ is fixed.

Note here equally the remark on the minimal value of $\varepsilon(N)$ as 6.

Proof. The rapid mixing part follows directly from Theorem B.10, which is implied by the MCMI decay and uniform local gap and an application of Pinsker's inequality. In detail, we get

$$\begin{aligned} \|\rho_t - \sigma\|_1 &\leq \sqrt{2D(\rho_t\|\sigma)} \leq \sqrt{2e^{-\alpha(\epsilon)t}D(\rho\|\sigma) + \epsilon} \\ &\leq \sqrt{2e^{-\alpha(\epsilon)t}N(2\beta gJ + \log d)) + \epsilon}. \end{aligned}$$

Hence, for times t larger than

$$\begin{aligned}
&\frac{1}{\alpha(\epsilon)}\log\frac{2(2\beta gJ+\log d)N}{\epsilon}\\
&\quad=\frac{D+1}{C}(5+4\beta gJ+8\log d)\left(2D\left(r+\xi\log\frac{KD^2DN}{\epsilon}\right)\right)^{D(1+\mu)}\\
&\quad\quad\log\frac{2(2\beta gJ+\log d)N}{\epsilon}\\
&\quad=\mathcal{O}\left(\log\left(\frac{N}{\epsilon}\right)^{1+D(1+\mu)}\right),
\end{aligned}$$

it holds that

$$\|\rho_t-\sigma\|_1\leq\sqrt{2\epsilon}.$$

Since this time also upper bounds the mixing time, replacing ϵ by $\frac{\varepsilon^2}{2}$ yields the first claim.

For the W_1 mixing proceed just as in the proof of Theorem C.1, but with Theorem B.10 instead of Theorem B.8. So fix a valid coarse-graining with constants (k,c,ℓ) and consider the weak TC from Theorem B.5 w.r.t this coarse-graining.

$$\begin{aligned}
\|\rho_t-\sigma\|_{W_1}&\leq\max_{a,i}2\sqrt{2}|C_{a,i}|\sqrt{ND(D+1)D(\rho_t\|\sigma)}+N\sqrt{2KD2^DN}e^{-\frac{c}{2\xi}}\\
&\leq 2\sqrt{2(D+1)}\ell^D\sqrt{N\left(e^{-\alpha(N\ell^{-2D}\epsilon)t}N(2\beta gJ+\log d)+\frac{N}{\ell^{2D}}\epsilon\right)}\\
&\quad+N\sqrt{N}\sqrt{2D2^DK}e^{-\frac{c}{\xi}}.
\end{aligned}$$

Now setting $k=r, c=\xi\log\frac{2D2^DKN}{\delta^2}=\mathcal{O}\left(\frac{N}{\delta^2}\right)$ and $l=2D(r+c)+1=\mathcal{O}\left(\frac{N}{\delta^2}\right)$ yields a valid coarse-graining and it is easily checked that for times t larger than

$$\frac{1}{\alpha(N\ell^{-2D}\epsilon)}\log\left(\frac{2\beta gJ+\log d}{\epsilon}\ell^{2D}\right)=\mathcal{O}\left(\left(\log\left(\frac{1}{\epsilon}\right)\right)^{1+D(1+\mu)}\right),$$

it holds that

$$\|\rho_t-\sigma\|_{W_1}\leq 2\sqrt{2(D+1)}N\sqrt{2\epsilon}+N\delta.$$

Setting $\epsilon=\frac{\delta^2}{16(D+1)}$ renaming $\delta=\frac{\varepsilon}{2}$ finishes the proof with the in the theorem claimed mixing time. In the first case, the requirement on the minimal size of ε comes from the one in Theorem B.10 rewritten in terms of $N=(2L+1)^D$ and applied to $\varepsilon=\sqrt{2\epsilon}$. In the second case, this is analogous to the proof of Theorem C.1. □

We want to emphasize again that alternatively to the assumption of the existence of a uniform polynomial gap one can also ask for a very high temperature to again reduce the requirements only to the decay of the MCMI (see Remark 5)

We now briefly examine the relationship between mixing times in trace distance and normalized Wasserstein distance:

Remark 7 (Relation between $t^{W_1}_{\text{mix}}(\varepsilon)$ and $t^{1}_{\text{mix}}(\varepsilon)$). For a GNS-symmetric quantum Markov semigroup $\{e^{t\mathcal{L}}\}_{t\geq 0}$ on the system $\mathcal{H} = \bigotimes_{i\in\mathcal{I}}\mathcal{H}_i$, the corresponding mixing times satisfy the relation

$$t^{W_1}_{\text{mix}}(\varepsilon) \leq t^{1}_{\text{mix}}(\varepsilon) \leq t^{W_1}_{\text{mix}}\left(\frac{\varepsilon}{2|\mathcal{I}|}\right). \tag{78}$$

This follows directly from Appendix 23. Consequently, a trace distance mixing time estimate such as that in Theorem C.2 immediately yields a normalized Wasserstein mixing time estimate with the same constant ε, while the converse is not true. Since the subadditivity property $t^{W_1}_{\text{mix}}(\varepsilon^k) \leq k\, t^{W_1}_{\text{mix}}(\varepsilon)$ does not hold in general for $k \in \mathbb{N}$, obtaining an estimate for $t^{1}_{\text{mix}}(\varepsilon)$ from results such as Theorem C.1 requires using the explicit bound on the Wasserstein distance, yielding

$$t^{1}_{\text{mix}}(\varepsilon) \leq t^{W_1}_{\text{mix}}\left(\frac{\varepsilon}{2|\mathcal{I}|}\right) \leq \text{quasi-poly}(N^2\log N^2)\cdot \text{quasi-poly}\left(\tfrac{1}{\varepsilon^2}\,\log\tfrac{1}{\varepsilon^2}\right).$$

C.3. Quasi-optimal Gibbs State Preparation from MCMI Decay

To convert the prior established mixing time bounds of the Davies semigroup into efficiency results of preparing their fixed points, we require (Davies) Lindblad simulation theorems. These give explicit constructions of quantum circuits, with circuit complexity bounds, which approximate $e^{t\mathcal{L}}$ in diamond norm. Such suitable circuits are for example constructed in [26,36,69,80], with the latter two being the more efficient ones.

Specifically in [26, Theorem III.2], the authors construct a Lindblad simulation algorithm for which it was shown that the complexity of implementing $e^{t\mathcal{L}^D_\Lambda}$ in terms of two-qubit gates, ancilla qubits and block encodings of the dissipative part and Hamiltonian part (see [26]) scales linear up to polylogarithmic corrections in $\frac{Nt_{\text{mix}}}{\epsilon}$. Combining this Lindblad simulation algorithm we get the following main result.

Theorem C.3. (Quasi-optimal sampling from Gibbs states that satisfy MCMI decay) *Let σ be a Gibbs states of $(\kappa, r)-$local, commuting, J-bounded Hamiltonian on $\Lambda \subset \mathbb{Z}^D$ which satisfies uniform MCMI decay. Then there exists a quantum algorithm (circuit) that outputs an ϵ-close in normalized W_1-distance, state using*

1. $\mathcal{O}\left(\text{poly}\log\left(N\,\text{quasi-poly}\left(\frac{1}{\epsilon^2}\,\text{poly}\log\frac{N}{\epsilon^2}\right)\right)\right) = \mathcal{O}\left(\text{poly}\log N, \text{poly}\log\frac{1}{\epsilon^2}\right)$ *ancilla qubits,*
2. $\mathcal{O}\left(N\,\text{quasi-poly}\left(\frac{1}{\epsilon^2}\,\text{poly}\log\frac{N}{\epsilon^2}\right)\right) = \mathcal{O}\left(N\text{quasi-log}(N), \text{quasi-poly}(\epsilon^{-2})\right)$ *two-qubit gates, block encodings of the Hamiltonian H_Λ and the dissipative part of $\mathcal{L}^D_\Lambda$.*

Remark 8. Such algorithms are usually referred to as optimal if the complexity and/or runtime scales as $\mathcal{O}(N)$ up to polylogarithmic corrections. Since our scaling is $\mathcal{O}(N)$ up to quasi-logarithmic corrections, we call it *quasi-optimal*, since it still scales better than $\mathcal{O}(N^2)$. In [26], they also explicitly construct these block encodings for the dissipative part of $\mathcal{L}_\Lambda^D$ in terms of simpler gates and the jump operators $S_{\alpha,k}^\omega$. Note though that any efficient Lindbaldian simulation algorithm, such as the one from [69], may be used to construct an efficient algorithm to sample from such Gibbs states. Our rapid mixing results imply optimal efficient preparation, analogous to the above, yielding circuit complexities and runtimes of $\mathcal{O}(N)$ up to polylogarithmic corrections.

Proof. The above bounds of the number of ancilla quits and necessary gates follow directly from implementing a quantum circuit $\mathcal{C}_\epsilon$ that ϵ approximates $e^{t\mathcal{L}_\Lambda^D}$ in diamond norm and Theorem C.1. The former is done in, for example, [26, Theorem III.2] and gives the in the theorem mentioned bound when substituting their t for $\mathcal{O}(N)t_{\text{mix}}^{W_1}(\varepsilon)$, since they consider normalized Lindbaldians, see (1.10) in [26]. This suffices since any quantum circuit that is ϵ close in diamond norm to $e^{t\mathcal{L}}$ also is ϵ-close in stabilized $1 \to \tilde{W}_1$ norm, where $\tilde{W}_1$ denotes the normalized W_1 distance. That is, let ρ be the input of this circuit $\mathcal{C}_\epsilon$, then

$$\begin{aligned}\frac{1}{N}\|\mathcal{C}_\epsilon(\rho)-\sigma\|_{W_1} &\le \frac{1}{N}\|\mathcal{C}_\epsilon(\rho)-e^{t_{\text{mix}}^{W_1}(\varepsilon)\mathcal{L}_\Lambda^D}(\rho)\|_{W_1}+\frac{1}{N}\|e^{t_{\text{mix}}^{W_1}(\varepsilon)\mathcal{L}_\Lambda^D}(\rho)-\sigma\|_{W_1}\\ &\le \|\mathcal{C}_\epsilon(\rho)-e^{t_{\text{mix}}^{W_1}(\varepsilon)\mathcal{L}_\Lambda^D}(\rho)\|_1+\varepsilon\\ &\le \|\mathcal{C}_\epsilon-e^{t_{\text{mix}}^{W_1}(\varepsilon)\mathcal{L}_\Lambda^D}\|_\diamond\|\rho\|_1+\varepsilon\le\epsilon+\varepsilon.\end{aligned}$$

We conclude by setting $\epsilon=\varepsilon$ and rescaling. □

D. Examples

In this section, we study systems that exhibit MCMI decay, which consequently leads to quasi-rapid Wasserstein mixing, and under a uniform local polynomial gap condition further to hyperrapid Wasserstein mixing and rapid mixing. We begin by deriving MCMI decay from the existence of a strong effective Hamiltonian in Appendix D.1 that exhibits a uniform bound in interaction norm. Next, in Appendix D.2, we assume the system Hamiltonian is marginal commuting, which, based on a result from [14], directly implies the existence of a strong effective Hamiltonian at high temperatures with a specified decay rate. By applying the result from Appendix D.1, we thus establish MCMI decay at high temperatures. We know that this commuting marginal assumption is fulfilled for every Hamiltonian composed of commuting Pauli strings (e.g. the Toric code in any dimension).[3] Lastly, note that in all the following results, we will not restate that MCMI decay implies quasi-rapid Wasserstein mixing, which, under a uniform local polynomial gap, can be strengthened to hyperrapid Wasserstein mixing and supplemented by rapid mixing in trace distance. If wanted the constants of the MCMI decay of all results below could be inserted into the result in Appendices C.1 and C.2 to obtain explicit decay rates.

[3]This result is shown by Sebastian Stengele in some currently unpublished notes.

D.1. MCMI Decay from Strong Effective Hamiltonian

The following lemma adapts the proof of [14, Theorem 4.1] to show the uniform decay of the MCMI (c.f. (39)) given the existence of an effective Hamiltonian with a uniform bound on its interaction norm.

Lemma D.1 (MCMI decay from locality of effective Hamiltonian). *In the setting of Appendix A.3 assume the existence of a strong effective Hamiltonian for H_Λ, with a uniform bound on the interaction norm, i.e.*

$$\left|\left|\left|\tilde{H}^A\right|\right|\right|_\mu := \sup_{x\in\Lambda} \sum_{X\subseteq\Lambda:x\in X} \|\tilde{h}^A_X\|_\infty e^{\mu\,\mathrm{diam}(X)} \leq \Delta \tag{79}$$

for $\mu, \Delta \geq 0$ independent of $A \subseteq \Lambda$. Then for any partition $\Lambda = A \sqcup B \sqcup C \sqcup D$

$$||\boldsymbol{H}(A:C|D)_\sigma||_\infty \leq 4\min\{|A|,|C|\}\Delta e^{-\mu\,\mathrm{dist}(A,C)}\,. \tag{80}$$

Proof. Without loss of generality, we can assume that $|A| \leq |C|$. Inserting the definition of the MCMI and using 3. from the definition of a strong effective Hamiltonian in Appendix A.4.3 we get

$$\begin{aligned}
||\mathbf{H}(A:C|D)_\sigma||_\infty &= \Big\| \sum_{X\subseteq\Lambda} \tilde{h}^{ACD}_X + \tilde{h}^D_X - \tilde{h}^{AD}_X - \tilde{h}^{CD}_X \Big\|_\infty \\
&= \Big\| \sum_{\substack{X\subseteq\Lambda:\\ X\cap A\neq\emptyset, X\cap C\neq\emptyset}} \tilde{h}^{ACD}_X + \tilde{h}^D_X - \tilde{h}^{AD}_X - \tilde{h}^{CD}_X \Big\|_\infty \\
&\leq \sum_{\substack{X\subseteq\Lambda:\\ X\cap A\neq\emptyset, X\cap C\neq\emptyset}} \left\|\tilde{h}^{ACD}_X\right\|_\infty + \left\|\tilde{h}^{D}_X\right\|_\infty + \left\|\tilde{h}^{AD}_X\right\|_\infty + \left\|\tilde{h}^{CD}_X\right\|_\infty
\end{aligned}$$

The first equality stems from the fact that

$$\tilde{h}^{ACD}_X + \tilde{h}^D_X - \tilde{h}^{AD}_X - \tilde{h}^{CD}_X = 0$$

if $X \cap C = \emptyset$ or if $X \cap A = \emptyset$. Indeed, if $X \cap C = \emptyset$, then $X \cap ACD = X \cap AD$ and $X \cap CD = X \cap D$ which, respectively, give $\tilde{h}^{ACD}_X = \tilde{h}^{AD}_X$ and $\tilde{h}^{CD}_X = \tilde{h}^D_X$ by 2. of Appendix A.4.3. The argument is analogous for $X \cap A = \emptyset$ by symmetry in A and C. Using (79), we obtain

$$\begin{aligned}
||\mathbf{H}(A:C|D)_\sigma||_\infty \leq &\sum_{\substack{X\subseteq\Lambda:\\ X\cap A\neq\emptyset, X\cap C\neq\emptyset}} e^{-\mu\,\mathrm{diam}(X)} e^{\mu\,\mathrm{diam}(X)} \\
&\left(\left\|\tilde{h}^{ACD}_X\right\|_\infty + \left\|\tilde{h}^{D}_X\right\|_\infty + \left\|\tilde{h}^{AD}_X\right\|_\infty + \left\|\tilde{h}^{CD}_X\right\|_\infty\right) \\
&\leq \sum_{x\in A} e^{-\mu\,\mathrm{dist}(x,C)} \sum_{\substack{X\subseteq\Lambda:\\ x\in X, X\cap C\neq\emptyset}} e^{\mu\,\mathrm{diam}(X)} \left(\left\|\tilde{h}^{ACD}_X\right\|_\infty + \left\|\tilde{h}^{D}_X\right\|_\infty\right. \\
&\quad \left. + \left\|\tilde{h}^{AD}_X\right\|_\infty + \left\|\tilde{h}^{CD}_X\right\|_\infty\right) \\
&\leq 4\Delta|A|e^{-\mu\,\mathrm{dist}(A,C)},
\end{aligned}$$

proving the claim. □

D.2. MCMI Decay from Commuting Marginals at High Temperature

Assuming that the system Hamiltonian H_Λ is marginal commuting now immediately gives an explicit uniform decay of the MCMI going through the result of [14, Theorem 3.8]. This is detailed in the theorem below.

Theorem D.2 (MCMI decay for marginal commuting systems at high temperature). *In the setting of Appendix A.3, assume that H_Λ is marginal commuting (c.f Appendix A.4.3 for a definition). Then, for $\beta < \frac{1}{\kappa g(1+\kappa g)e^2\, gJ}$ and for every partition $\Lambda = A \sqcup B \sqcup C \sqcup D$, we have*

$$||\boldsymbol{H}_\sigma(A:C|D)||_\infty \leq 4\min\{|A|,|C|\}e^{-\mu\,\mathrm{dist}(A,C)}$$

with $\mu = \frac{1}{r}\log\left(\frac{1}{g\kappa(1+g\kappa)e^2 gJ\beta}\right)$.

Proof. Assume that $\beta < \frac{1}{\kappa g(1+\kappa g)e^2 gJ}$ and set $\mu_\epsilon = \frac{1+\epsilon}{r}\log\left(\frac{1}{g\kappa(1+g\kappa)e^2\, gJ\beta}\right) > 0$ for $\epsilon > 0$. Then

$$|||H_\Lambda|||_{\mu_\epsilon} \leq gJe^{\mu_\epsilon r} < \frac{1}{g\kappa(1+g\kappa)e^2\beta}\,.$$

Reordering the above gives

$$\beta < \frac{1}{g\kappa(1+g\kappa)e^2\,|||H_\Lambda|||_{\mu_\epsilon}} \leq \frac{1}{\mathfrak{d}(1+\mathfrak{d})e^2\,|||H_\Lambda|||_{\mu_\epsilon}}\,. \tag{81}$$

In the last inequality, we used $\mathfrak{d} \leq g\kappa$, where $\mathfrak{d}$ is the degree of the interaction graph (see [14]). Combining this temperature constraint with the commutativity of the generated algebra and its preservation under partial traces, all conditions of [14, Theorem 3.8] are satisfied. We can therefore conclude the existence of a strong effective Hamiltonian with decay

$$\left|\left|\left|\tilde{H}^A\right|\right|\right|_{\mu_\epsilon} \leq 1$$

for every $A \subseteq \Lambda$. Now, by Lemma D.1, we immediately conclude the claim with decay rate μ_ϵ. Taking $\epsilon \to 0$ then concludes the claim with the stated decay rate. $\square$

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